

**DEPARTMENT OF COMPUTER SCIENCE AND
ENGINEERING**

Digital Forensics Laboratory Course Record

DIGITAL FORENSICS LABORATORY

Practical Laboratory Observations, Evidence Acquisitions & Forensic Analysis

Laboratory Record: Experiments 01 – 05

Academic Year: 2026 – 2027

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EXPERIMENT 01

Evidence Acquisition Using AccessData FTK Imager

OBJECTIVE

To acquire digital evidence in a forensically sound manner using AccessData FTK Imager, preserve evidentiary integrity, calculate cryptographic hashes (MD5 and SHA-1), and verify image authenticity against source data.

TOOLS / REQUIREMENTS

- **Forensic Software:** AccessData FTK Imager / Exterro FTK Imager (Version 8.3.0.27)
- **Host Operating System:** Microsoft Windows 11 Home (Build 6.3.26200)
- **Forensic Environment:** Write-blocking methodology / Read-only evidence handling
- **Evidence Storage:** Local directory structure for source and forensic image destination

EXPERIMENT SCENARIO

Controlled laboratory evidence was used for this experiment.

A simulated digital forensic acquisition scenario was established to demonstrate the acquisition and verification of digital evidence. The investigator was tasked with importing targeted laboratory evidence items, recording formal case metadata, creating an evidence container (.ad1 custom content image format), and validating evidence integrity via cryptographic hashing.

EVIDENCE / INPUT

- **Source Evidence Path:** C:\DFIR-Lab\EXP-03\Source-Evidence
- **Case Identifier:** CASE-2026-DF-003
- **Evidence Identifier:** EVID-001
- **Investigator / Examiner:** SEKHAR REDDY
- **Evidence Description:** Controlled Laboratory Test Evidence
- **Target Files Acquired:**
 - case_notes.txt (232 bytes / 1 KB)
 - employee_list.txt (104 bytes / 1 KB)
 - investigation_log.txt (199 bytes / 1 KB)

PROCEDURE

Step 1 — Adding Evidence Item and Previewing Case Notes

The FTK Imager application was launched and the source evidence folder was mounted as a custom content source. The file `case_notes.txt` was selected in the file list to preview its contents and verify evidentiary details in the lower viewer pane.

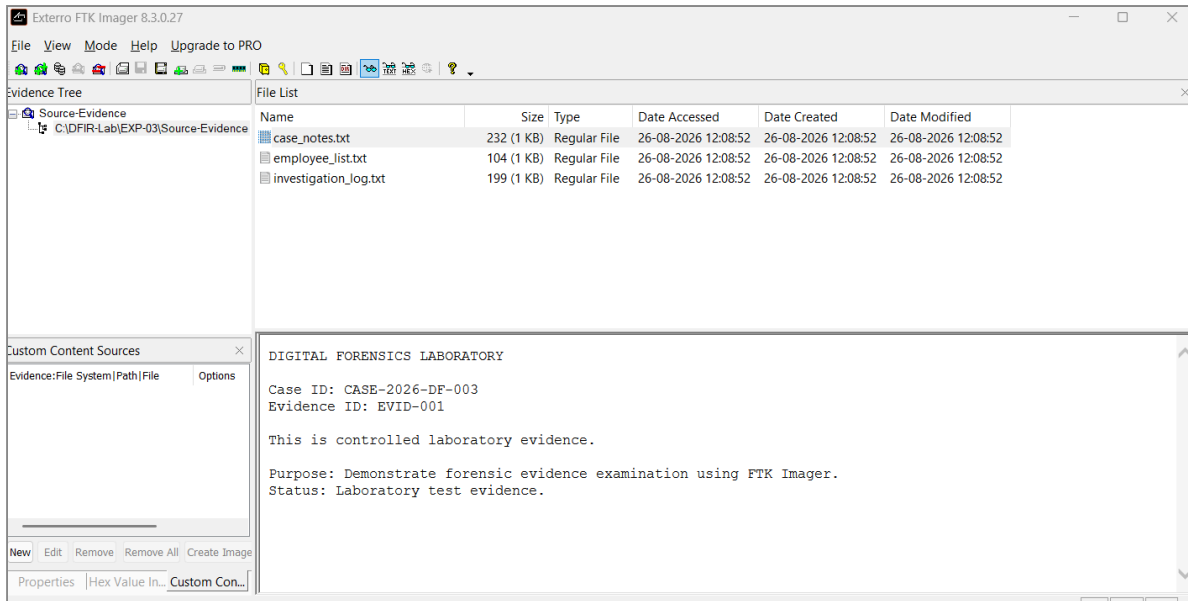


Figure 1.1: Evidence loaded and case notes preview in FTK Imager

Step 2 — Examining Evidence Content — Employee List

The file `employee_list.txt` was selected from the file list to inspect its contents. The text viewer displayed records of test employee IDs, user roles, and assigned laboratory departments (EMP001, Test User, Security; EMP002, Lab User, Forensics; EMP003, Demo User, IT).

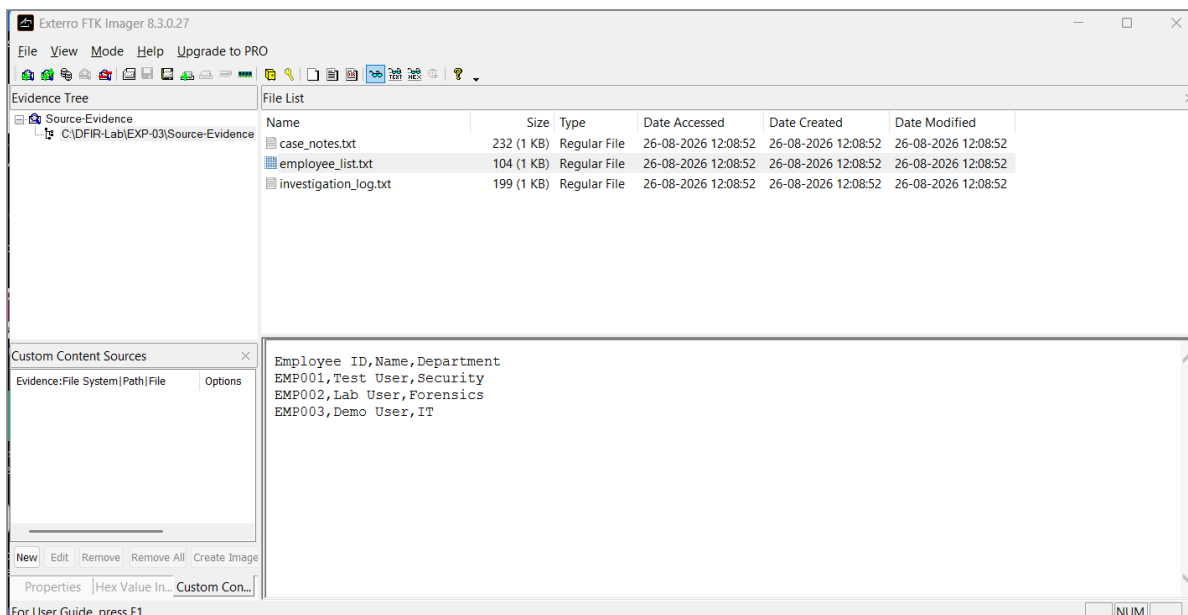


Figure 1.2: Employee list preview in FTK Imager

Step 3 — Examining Evidence Content — Investigation Log

The file `investigation_log.txt` was inspected to confirm timestamps and event records created during the controlled test setup.

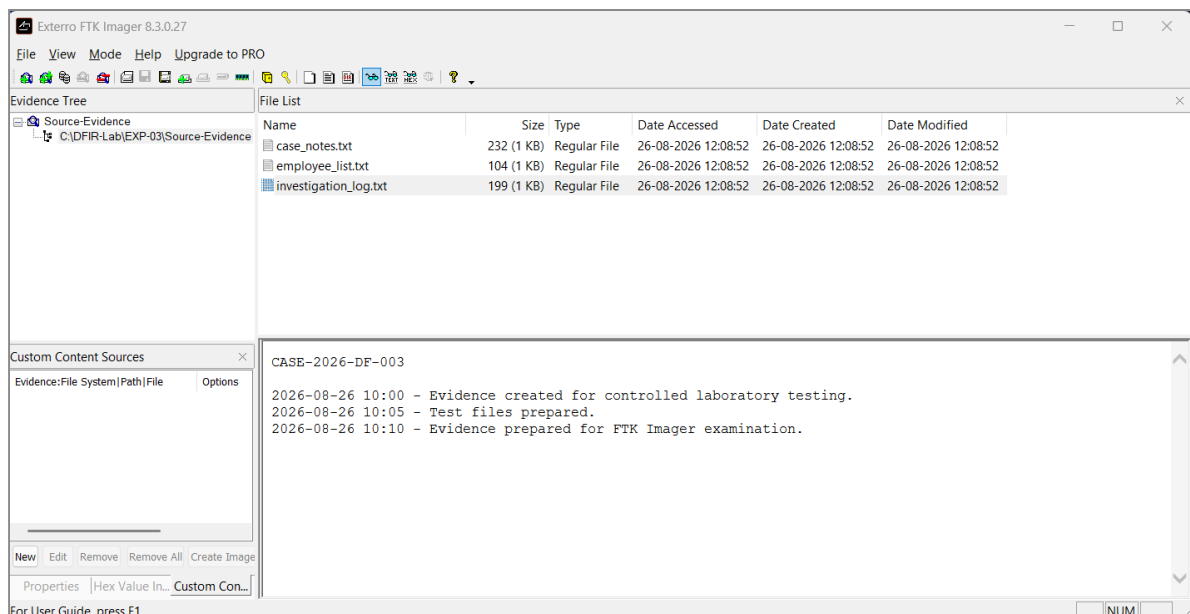


Figure 1.3: Investigation log preview in FTK Imager

Step 4 — Creating Forensic Image and Entering Case Metadata

The *Create Image* function was initiated. In the **Evidence Item Information** dialog, official forensic case metadata was entered:

- **Case Number:** CASE-2026-DF-003
- **Evidence Number:** EVID-001
- **Unique Description:** Controlled Laboratory Test Evidence
- **Examiner:** SEKHAR REDDY
- **Notes:** [None]

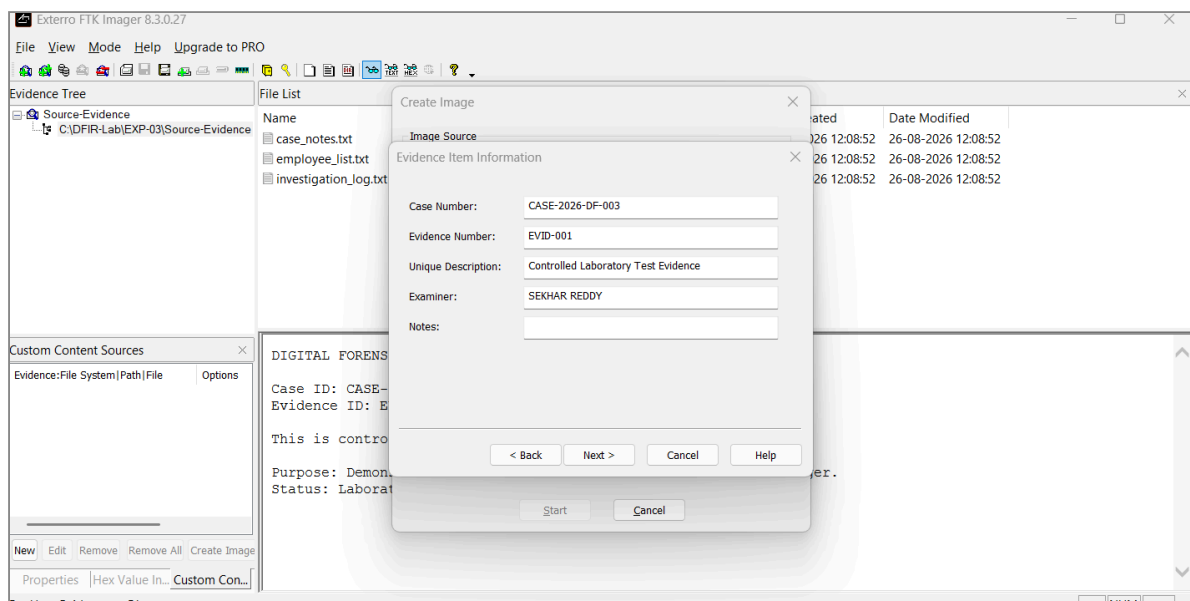


Figure 1.4: Evidence item information and case metadata entry

Step 5 — Forensic Image Verification Results

The evidence acquisition and post-creation verification processes were executed. The **Drive/Image Verify Results** window confirmed successful completion:

Parameter	Value	Status
Target Image	E01.ad1	Created
MD5 Computed Hash	187811d9c772f8ad92fe05384d58d2bf	Match
MD5 Report Hash	187811d9c772f8ad92fe05384d58d2bf	Verified
SHA-1 Computed Hash	512682cfc03b0d41085ee28c1716bdbfcd5912f9	Match
SHA-1 Report Hash	512682cfc03b0d41085ee28c1716bdbfcd5912f9	Verified

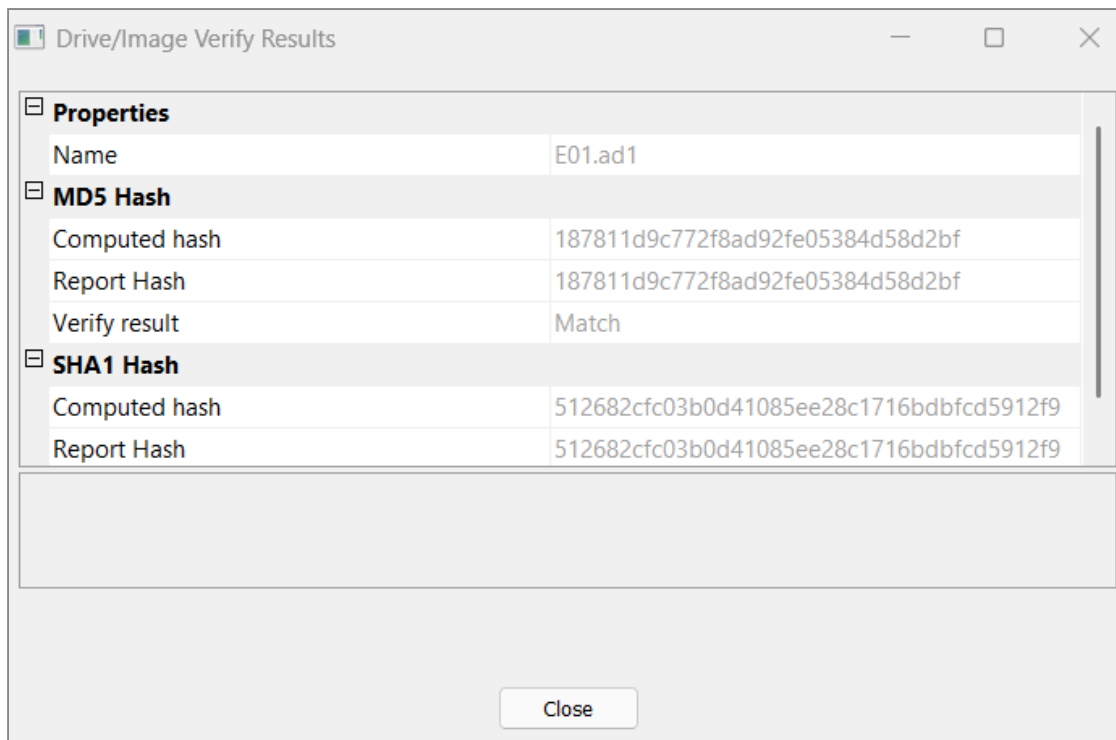


Figure 1.5: Drive/image verify results showing MD5 and SHA-1 match

Step 6 — Reviewing Acquired Forensic Image in Evidence Tree

The generated forensic image directory C:\DFIR-Lab\EXP-03\Forensic-Image was added to the FTK Imager evidence tree. The folder structure revealed:

- E01.ad1 (Forensic Image container — 3,118 bytes)
- E01.ad1.txt (Forensic Acquisition and Verification Log — 1,796 bytes)

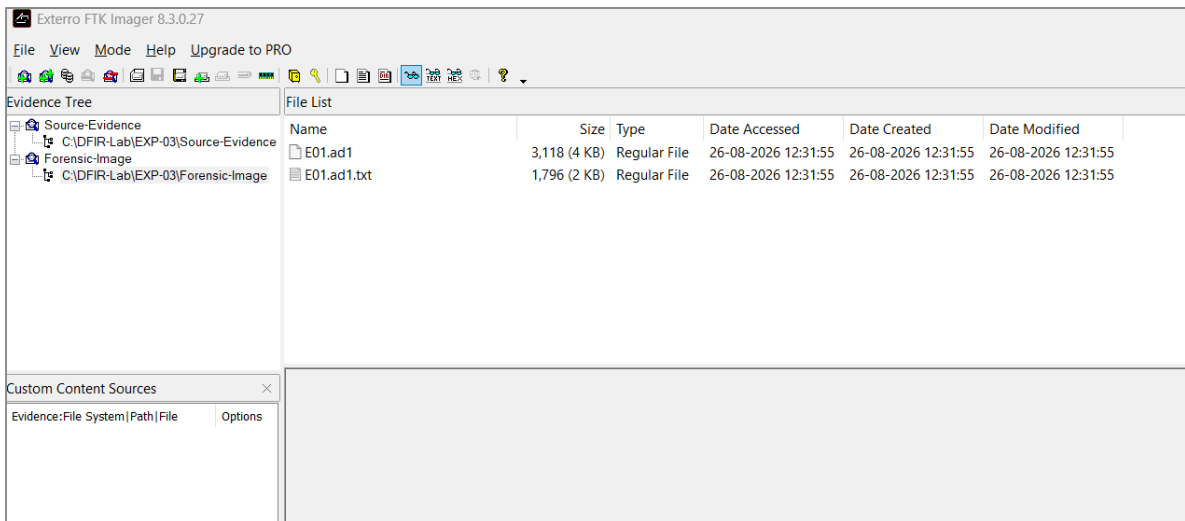


Figure 1.6: Evidence tree displaying acquired forensic image E01.ad1

Step 7 — Examining Acquisition and Verification Summary Report

The generated acquisition summary report E01.ad1.txt was loaded and examined in FTK Imager. The log documented complete hardware and environment details (Host: SURYA, Windows 11 Build 6.3.26200, Intel Core i7-13620H, 16 GB RAM), timestamps (Wed Aug 26 18:01:55 2026), and verified MD5 and SHA-1 hash calculations.

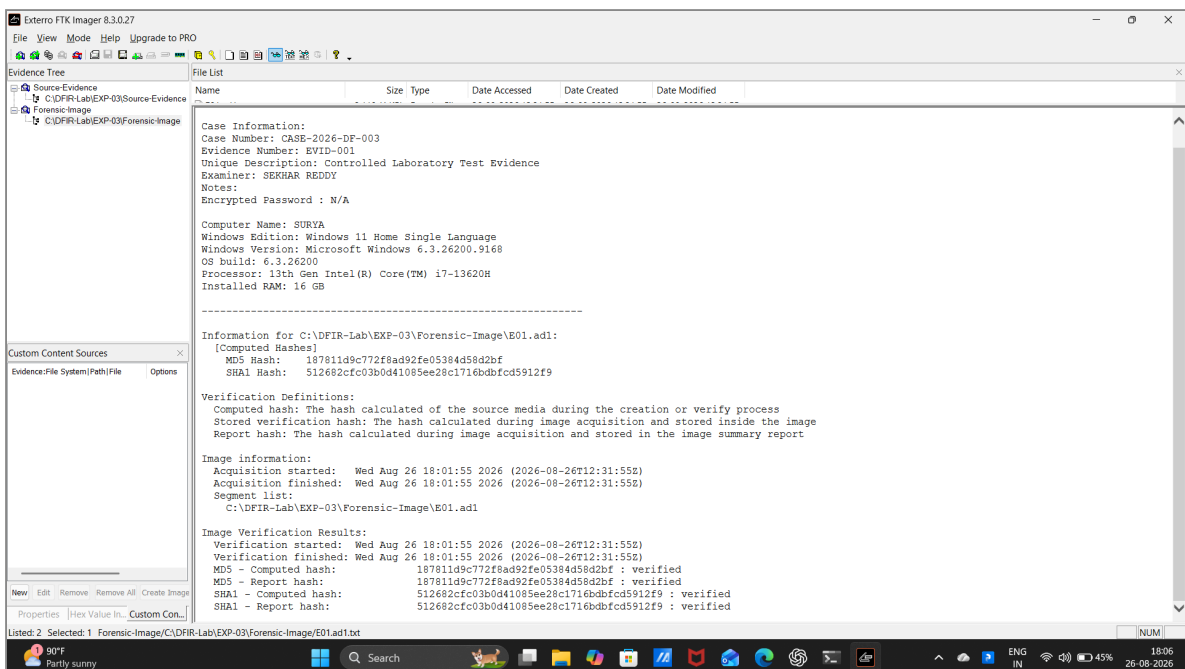


Figure 1.7: Acquisition and verification summary log report

OBSERVATIONS

1. The source files were loaded into FTK Imager without altering file attributes or timestamps.

2. The image creation wizard allowed input of case metadata (CASE-2026-DF-003, Examiner: SEKHAR REDDY).
3. An .ad1 evidence container (E01.ad1) and its associated log (E01.ad1.txt) were successfully created in C:\DFIR-Lab\EXP-03\Forensic-Image.
4. Cryptographic MD5 hash computed was 187811d9c772f8ad92fe05384d58d2bf (Match).
5. Cryptographic SHA-1 hash computed was 512682cfc03b0d41085ee28c1716bdbfcd5912f9 (Match).

FINDINGS

- **Data Integrity Assurance:** The computed MD5 and SHA-1 hashes of the acquired image match the reported source hashes exactly, proving that no data modification or corruption occurred during acquisition.
- **Forensic Soundness:** The use of structured metadata and automated verification logs establishes a defensible chain of custody and forensic record for the acquired container.

RUBRICS

Criteria & Marks Assigned	Mark Allotted	Mark Awarded
1. GitHub Activity & Submission Regularity	3	
2. Application of Forensic Tools & Practical Execution	3	
3. Documentation & Reporting	2	
4. Engagement, Problem-Solving & Team Collaboration	2	
Total	10	

RESULT

Successfully Completed. The evidence acquisition process was executed in AccessData/Exterro FTK Imager with complete case metadata and verified MD5/SHA-1 cryptographic integrity.

CONCLUSION

FTK Imager successfully acquired the designated laboratory evidence into a forensically sound image format (.ad1). Both MD5 and SHA-1 verification hashes confirmed bitstream integrity and preservation of evidence.

REFERENCES

- AccessData FTK Imager User Guide
- Digital Forensics Laboratory Manual: Evidence Acquisition Using AccessData FTK Imager

EXPERIMENT 02

Recover Deleted or Damaged Files from a Storage Device Using TestDisk

OBJECTIVE

To recover deleted or damaged files from a storage device, analyze partition table structures, and restore lost data to a secure location using TestDisk.

TOOLS / REQUIREMENTS

- **Forensic Software:** TestDisk 7.3-WIP (Data Recovery Utility, Christophe GRENIER / CGSecurity)
- **Host Operating System:** Microsoft Windows 11
- **Storage Disk / Target Volume:** Virtual storage disk volume (\\.\PhysicalDrive1 — 2147 MB / 2048 MiB NTFS volume [New Volume])

EXPERIMENT SCENARIO

Controlled laboratory evidence was used for this experiment.

A storage volume was configured with a deleted test credential document (User_credentials.txt.txt). The forensic investigator was tasked with launching TestDisk, evaluating the partition table architecture, analyzing the volume structure, locating deleted file records from the Master File Table / directory index, and recovering the deleted artifact to a designated forensic directory without modifying the original evidence.

EVIDENCE / INPUT

- **Target Storage Volume:** Disk \\.\PhysicalDrive1 - 2147 MB / 2048 MiB - Msft Virtual Disk
- **Partition Structure:** 1 P HPFS - NTFS (CHS 261 255 63, Sector size 512, [New Volume])
- **Target Deleted Artifact:** ./User_credentials.txt.txt (Size: 18 bytes, Deleted Timestamp: 26-Aug-2026 19:29)
- **Recovery Destination Path:** C:\DFIR-Lab\

PROCEDURE

Step 1 — TestDisk Initialization and Log File Creation

The TestDisk executable (testdisk_win.exe) was launched with administrative privileges. The option [Create] Create a new log file was selected to record diagnostic messages and recovery actions in testdisk.log.

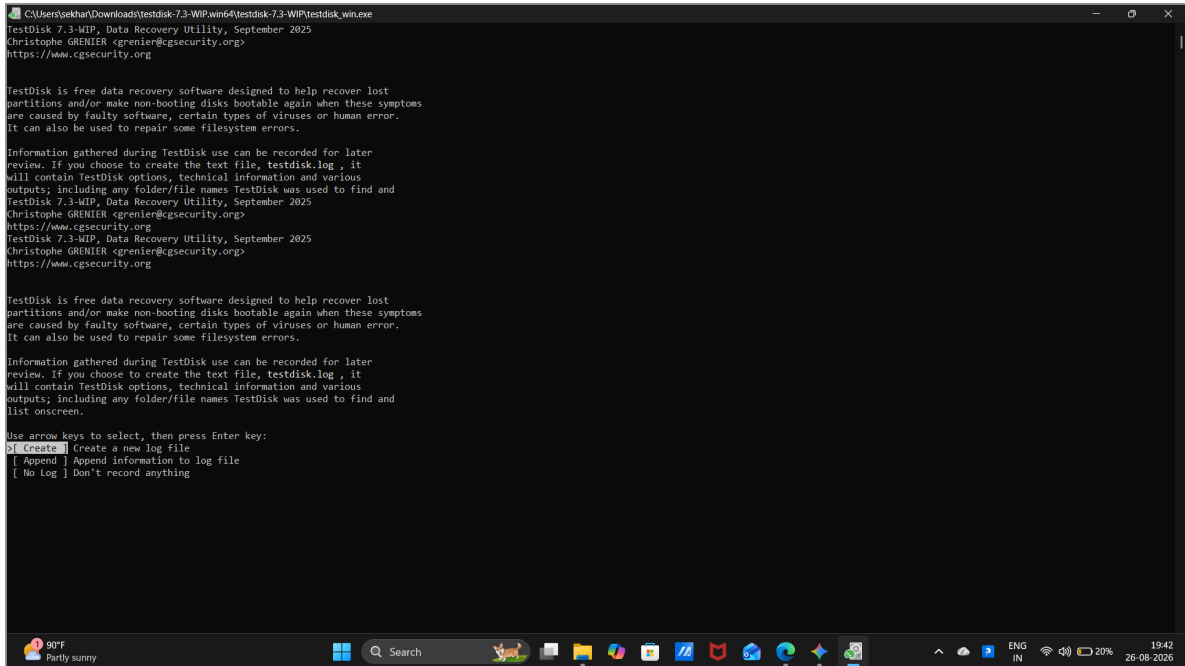


Figure 2.1: TestDisk startup and log file creation prompt

Step 2 — Storage Device and Partition Table Type Selection

The target virtual storage disk Disk \\.\PhysicalDrive1 (Msft Virtual Disk, 2048 MiB) was selected. The partition table type was set to [Intel] Intel/PC partition based on automated architecture detection.

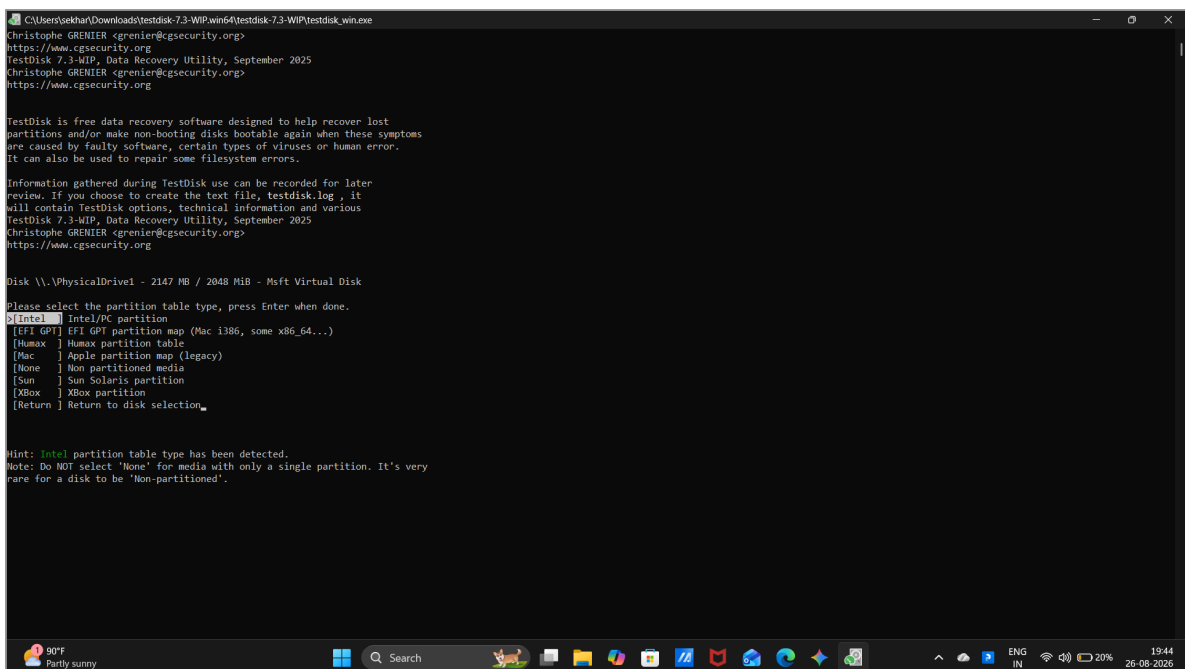


Figure 2.2: Storage device and Intel/PC partition table type selection

Step 3 — Current Partition Structure Analysis

The [Analyse] function was selected to inspect the volume geometry, check partition boundaries, and search for lost partitions.

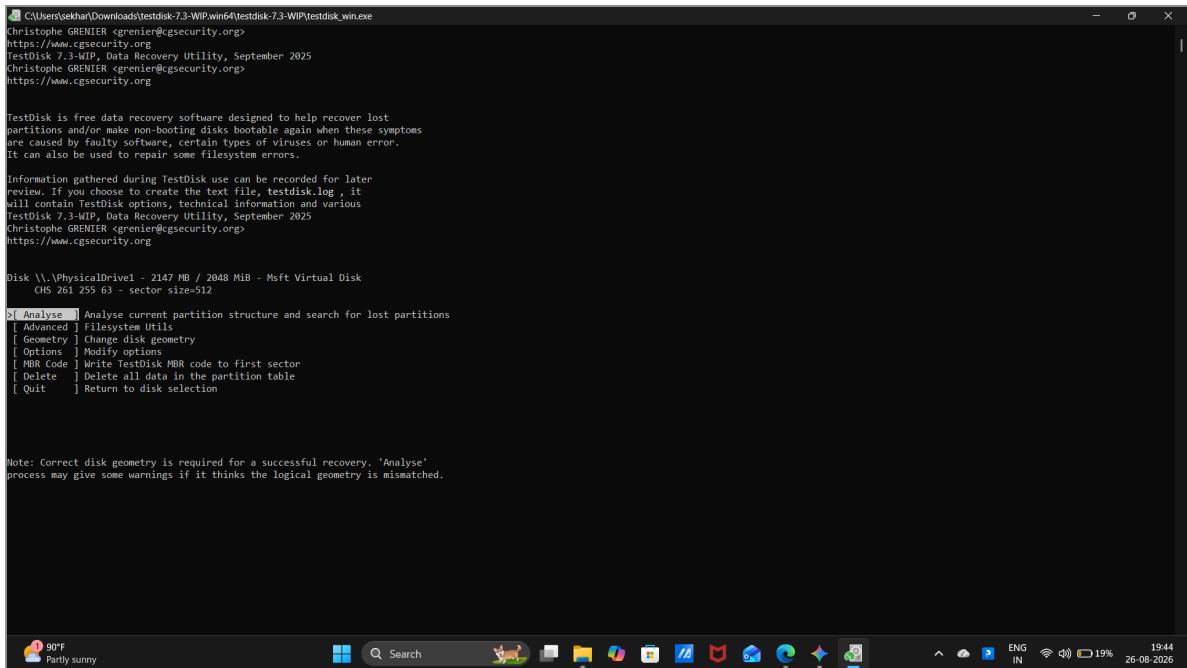


Figure 2.3: Current partition structure analysis

Step 4 — Navigating Advanced Filesystem Utilities

The [Advanced] Filesystem Utils menu was accessed to interact with filesystem-specific recovery mechanisms for the NTFS partition.

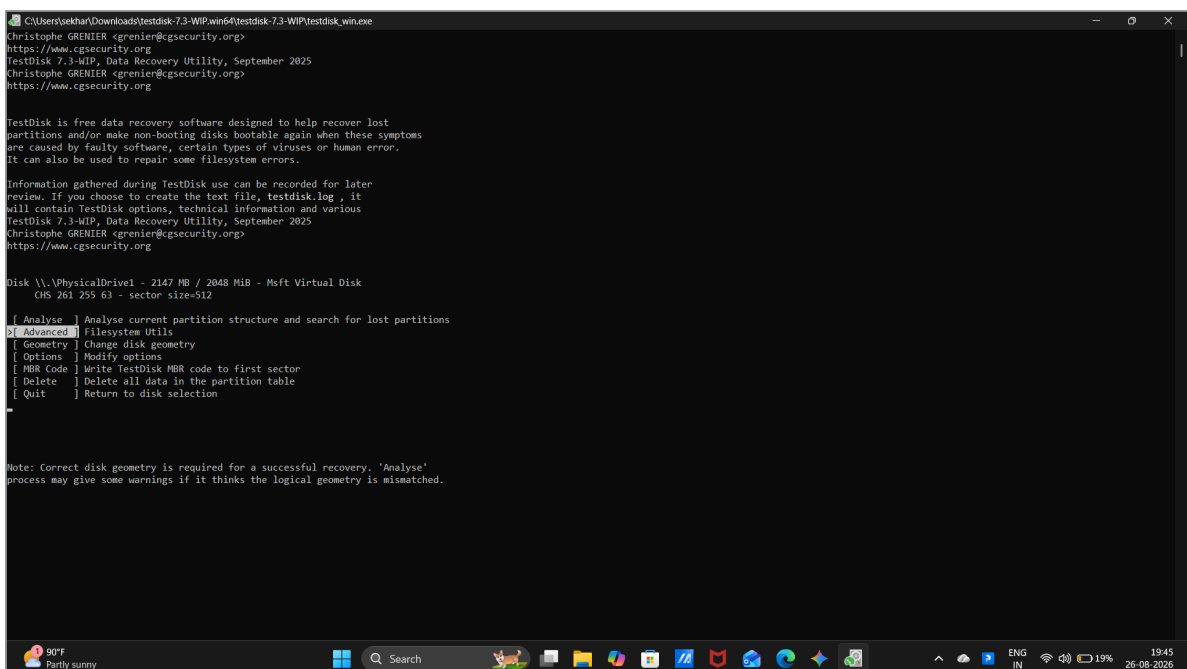


Figure 2.4: Advanced filesystem utilities menu

Step 5 — Inspecting Deleted Files on NTFS Volume

The undelete utility parsed the directory index and listed deleted file entries on [New Volume]. The deleted file `./User_credentials.txt.txt` (18 bytes, deleted 26-Aug-2026 19:29) was discovered.

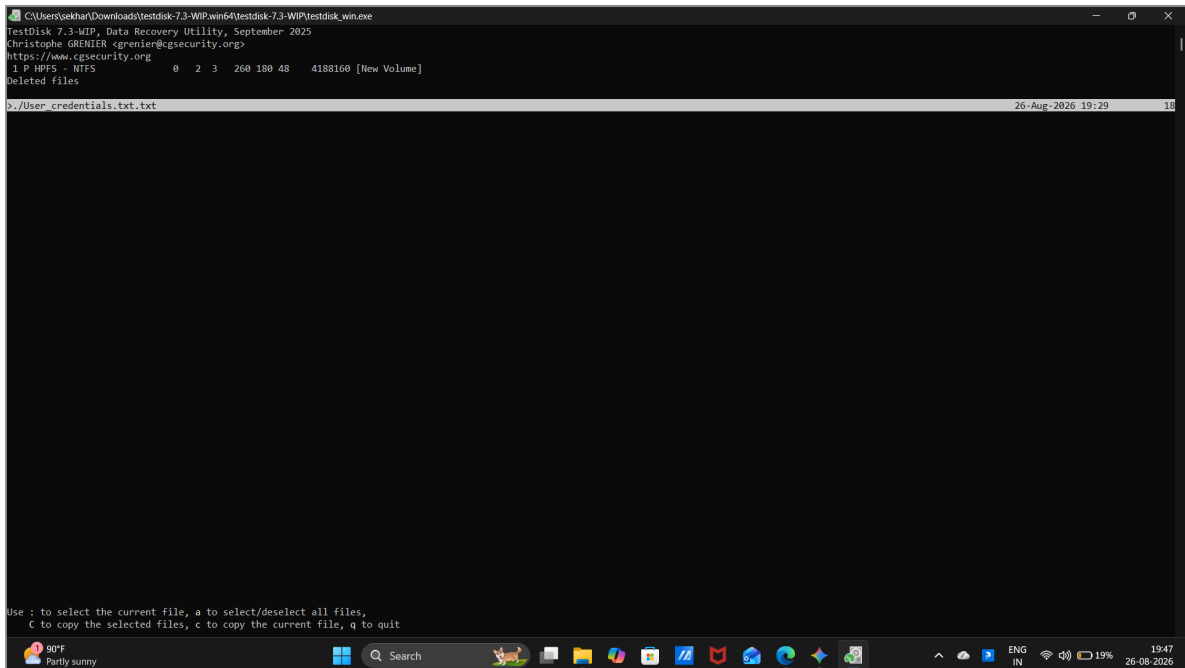


Figure 2.5: Deleted files listing on NTFS volume

Step 6 — Selecting File for Extraction and Recovery

The target artifact `./User_credentials.txt.txt` was selected and highlighted in green to prepare for copying out of unallocated space.

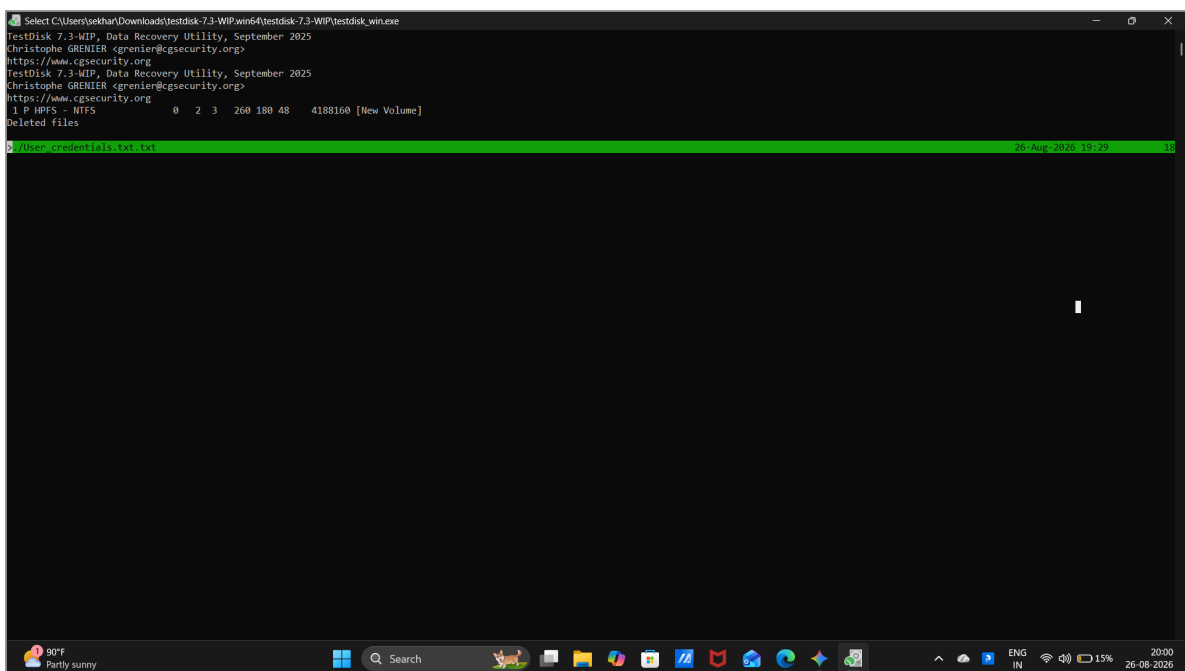


Figure 2.6: Selection and highlighting of deleted file for recovery

Step 7 — Choosing Secure Recovery Destination Directory

The destination drive path `c` was chosen to safely write out the recovered file outside the source storage volume.

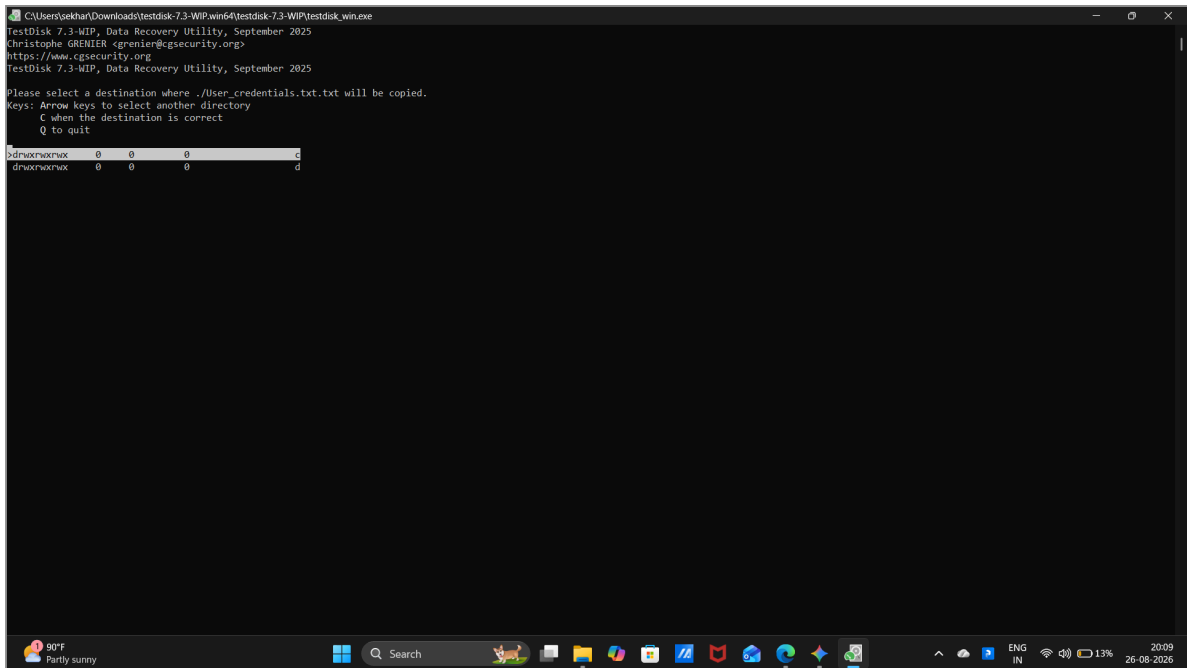


Figure 2.7: Selection of recovery destination directory

Step 8 — Verifying Recovered File in File Explorer

Windows File Explorer was opened to confirm successful restoration. The recovered file `User_credentials.txt` (1 KB, modified 26-08-2026 19:29) was verified inside `C:\DFIR-Lab\`.

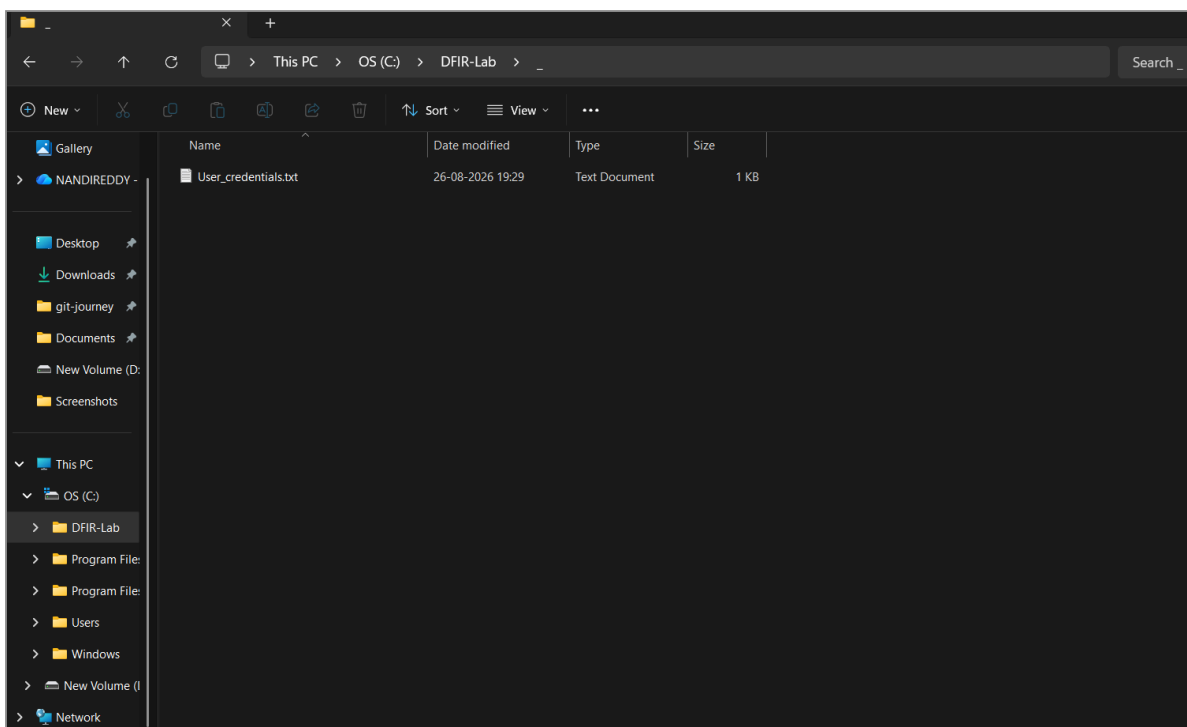


Figure 2.8: Verification of recovered file in Windows File Explorer

OBSERVATIONS

1. TestDisk accurately identified the virtual storage disk `\\.\PhysicalDrive1` (2048 MiB).

2. The NTFS volume structure was recognized at CHS 261 255 63 ([New Volume]).
3. The deleted file entry `./User_credentials.txt.txt` was discovered with intact metadata (18 bytes).
4. The file was copied to `C:\DFIR-Lab\` and verified in File Explorer.

FINDINGS

- **MFT Record Preservation:** The NTFS file record remained intact in the Master File Table / directory index prior to cluster reallocation.
- **Forensic Recovery Integrity:** TestDisk successfully extracted and restored the deleted text file without altering the source volume structure.

RUBRICS

Criteria & Marks Assigned	Mark Allotted	Mark Awarded
1. GitHub Activity & Submission Regularity	3	
2. Application of Forensic Tools & Practical Execution	3	
3. Documentation & Reporting	2	
4. Engagement, Problem-Solving & Team Collaboration	2	
Total	10	

RESULT

Successfully Completed. The deleted credential artifact was located on the NTFS storage volume, extracted, and successfully restored to `C:\DFIR-Lab\`.

CONCLUSION

TestDisk successfully identified the partition structure, scanned for deleted records, and recovered the lost text document to the forensic directory.

REFERENCES

- TestDisk Documentation (Christophe GRENIER / CGSecurity)
- Digital Forensics Laboratory Manual: Recover deleted or damaged files from a storage device using Test Disk

EXPERIMENT 03

Password Capturing Using Wireshark

OBJECTIVE

To capture and analyze network traffic using Wireshark network protocol analyzer, inspect communication protocols, evaluate packet headers across network layers, and analyze data streams for credential exposure and encryption posture.

TOOLS / REQUIREMENTS

- **Forensic Software:** Wireshark Network Protocol Analyzer (Versions 4.6.4 and 4.6.8)
- **Operating Environments:** Linux Virtual Machine / Microsoft Windows 11
- **Network Interface:** Wireless Network Adapter (Wi-Fi / `enp0s3`)

EXPERIMENT SCENARIO

Controlled laboratory evidence was used for this experiment.

A packet sniffing and protocol analysis scenario was executed to evaluate network traffic on active interfaces. The investigator captured live packets, applied display filters (`dns`, `ip`, `tcp`, `udp`), analyzed deep frame headers across OSI layers, followed TCP streams, and examined the protocol distribution hierarchy.

EVIDENCE / INPUT

- **Capture Interface:** Wi-Fi (`\\Device\\NPF_{37BBBAAB-CB2A-4351-9D61-0CA28A47E4ED}`)
- **Local Host IP / MAC:** `192.168.1.37 / 58:02:05:4c:2f:b8`
- **Total Packets Captured:** 755 packets (536 TCP/TLS packets analyzed)
- **Target Protocols:** Ethernet II, IPv4, IPv6, TCP, UDP, DNS, TLSv1.2, SSDP

PROCEDURE

Step 1 — Wireshark Interface Selection in Linux Virtual Machine

Wireshark 4.6.4 was launched in a Linux virtual environment to examine available capture interfaces, highlighting network adapter `enp0s3`.

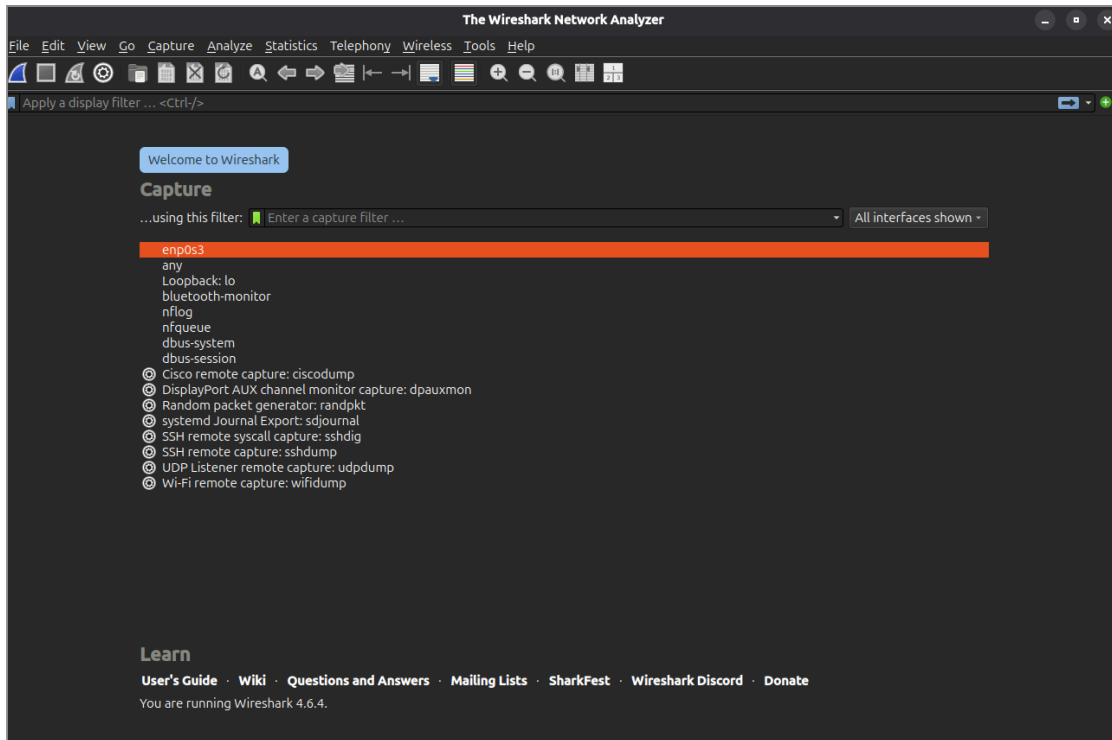


Figure 3.1: Wireshark capture interface selection in Linux VM

Step 2 — Wireshark Initialization on Windows Host

Wireshark 4.6.8 was opened on the Windows host, displaying active interface traffic activity graphs on the Wi-Fi adapter (IP 169.254.93.188 / `fe80::b38c:733c:d43:1aea`).

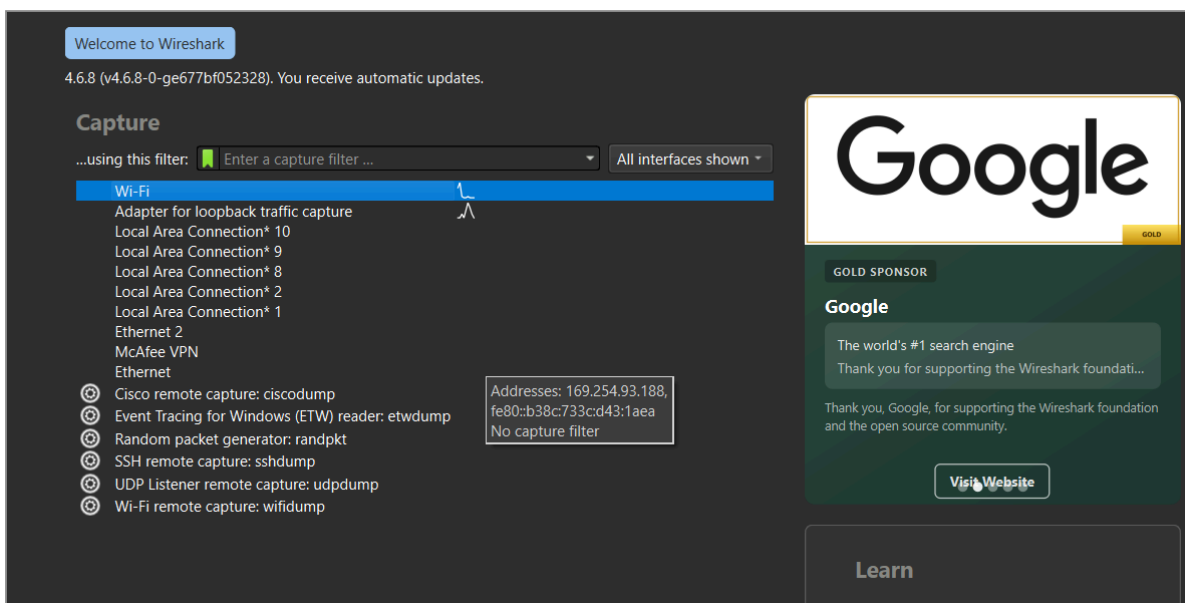


Figure 3.2: Wireshark welcome screen and Wi-Fi interface selection on Windows

Step 3 — Initiating Live Packet Capture on Wi-Fi Interface

Packet capturing was initiated on the Wi-Fi interface. Live network traffic involving local host 192.168.1.37 and external nodes was captured in real time.

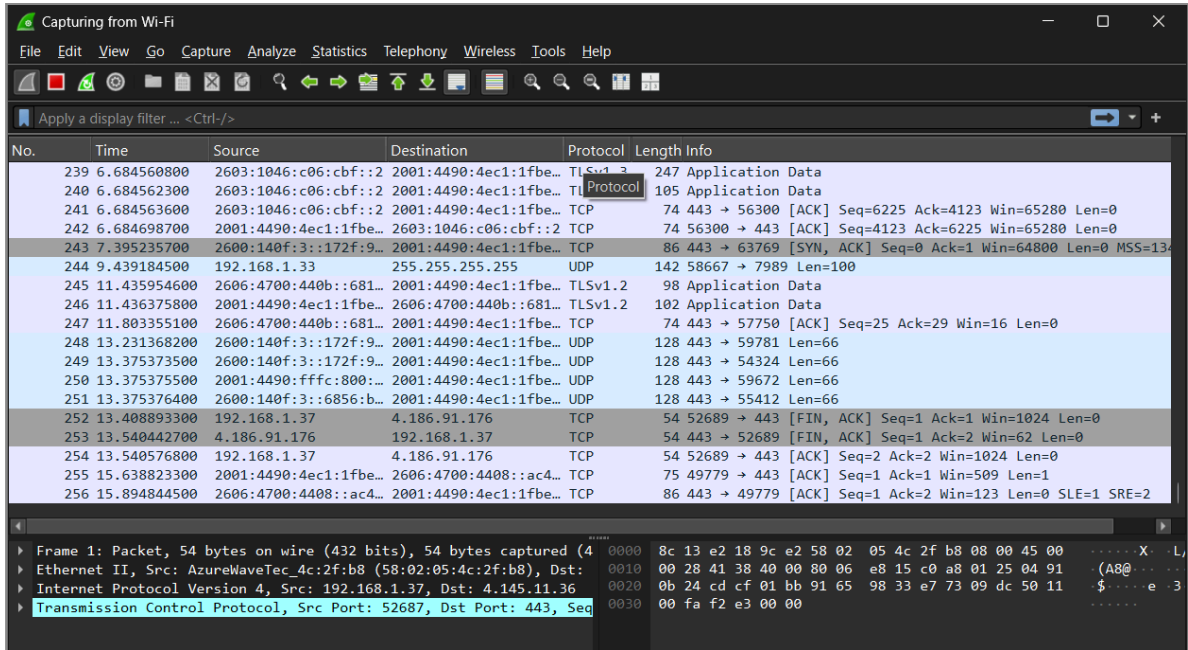


Figure 3.3: Live packet capture on Wi-Fi interface

Step 4 — Overview of Captured Transmission Frames

Captured packets were reviewed, showing standard TCP three-way handshake segments, TLSv1.2 Server Hello, Client Key Exchange, and connection termination flags (FIN/ACK, RST/ACK).

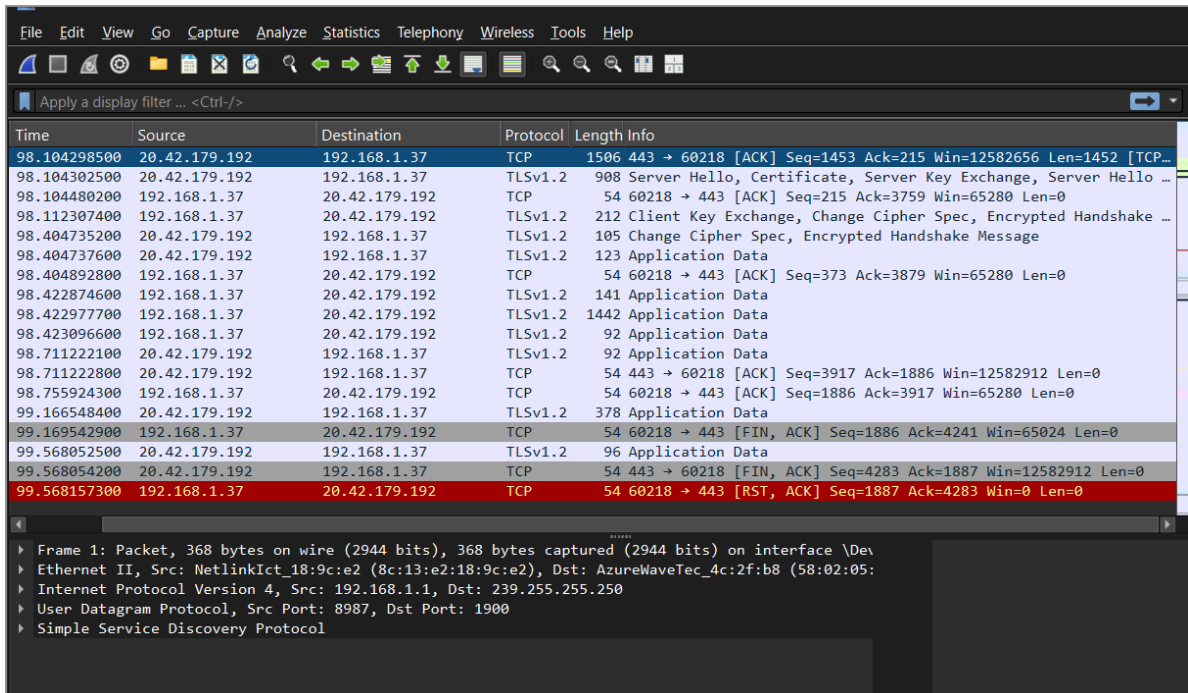


Figure 3.4: Overview of captured transmission frames

Step 5 — Filtering DNS Protocol Queries and Responses

The display filter dns was applied to isolate domain name resolution traffic, identifying standard queries and A/AAAA responses for google.com, chatgpt.com, and github.com.

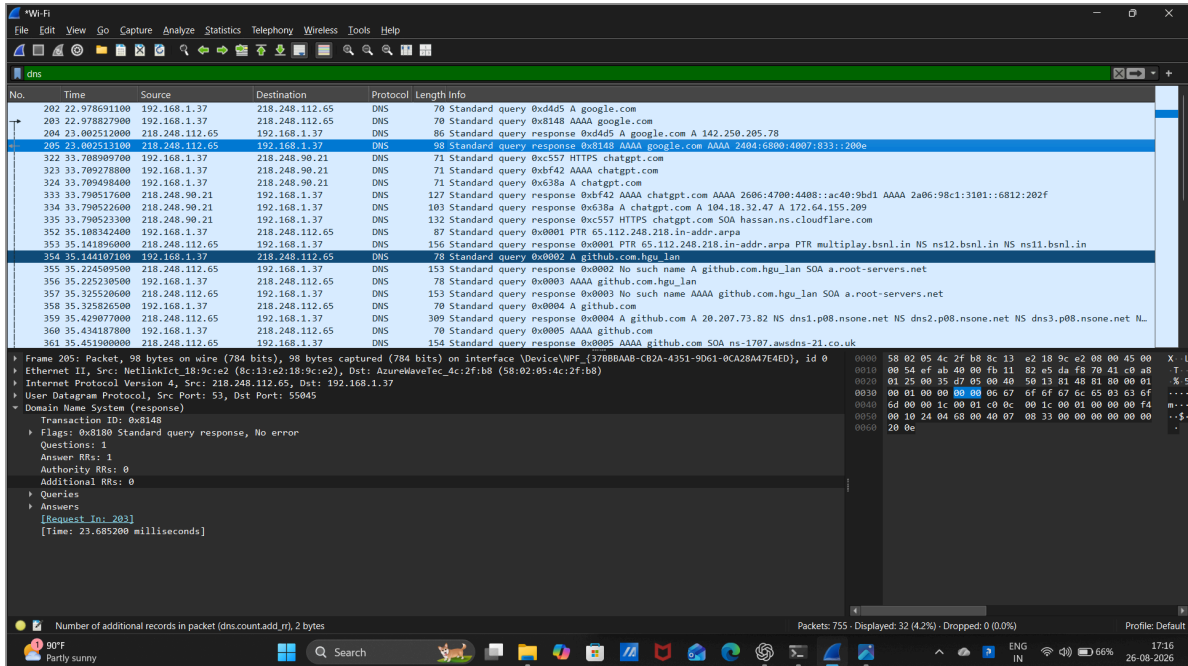


Figure 3.5: Applied DNS display filter showing domain queries

Step 6 — Filtering Internet Protocol (IPv4/IPv6) Traffic

The display filter ip was applied to filter network-layer packets between local gateway 192.168.1.1, host 192.168.1.37, and DNS resolvers (218.248.112.65).

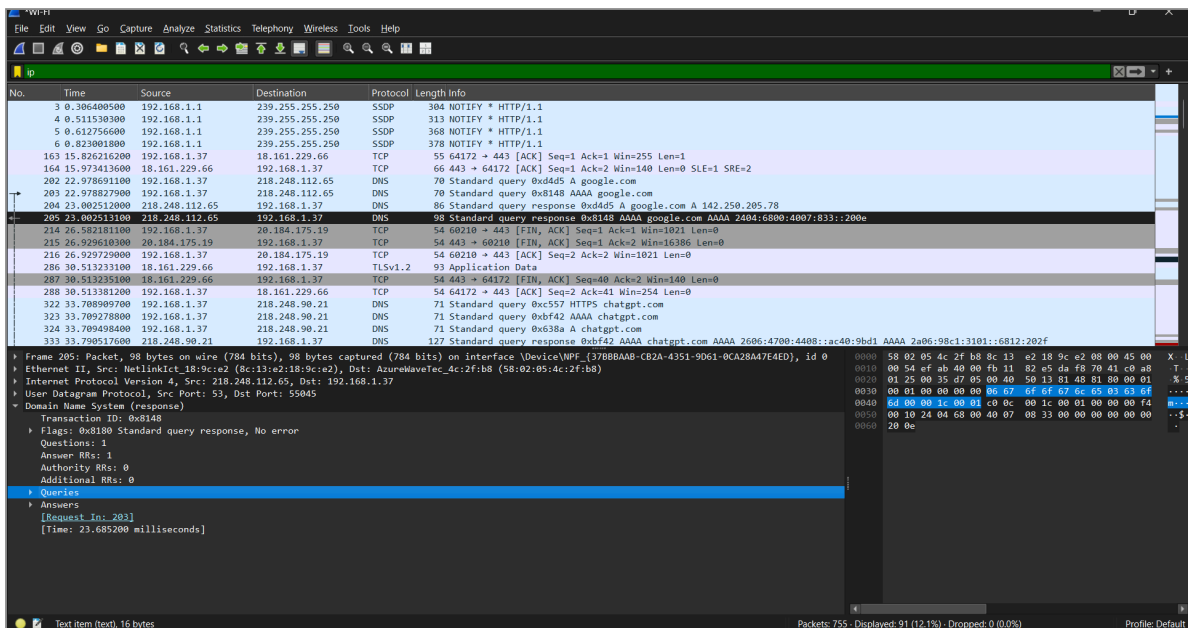


Figure 3.6: Applied IP display filter

Step 7 — Filtering Transmission Control Protocol (TCP) Streams

The display filter `tcp` was applied to analyze transport-layer communication streams, TCP sequence/acknowledgment numbers, and reassembled TCP PDUs.

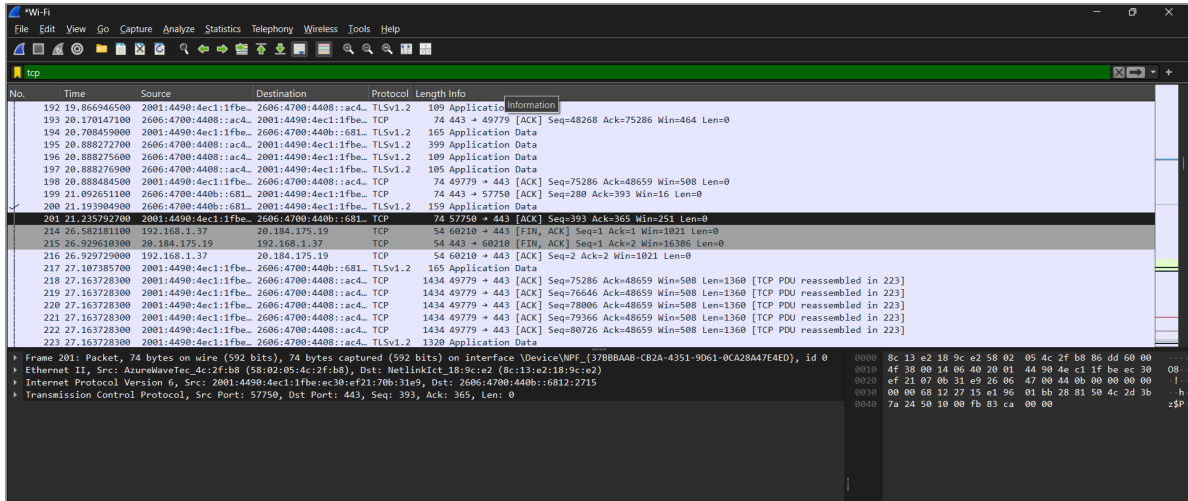


Figure 3.7: Applied TCP display filter

Step 8 — Filtering User Datagram Protocol (UDP) Traffic

The display filter `udp` was applied to inspect connectionless datagram traffic, revealing Simple Service Discovery Protocol (SSDP) multicast notifications on destination `239.255.255.250:1900`.

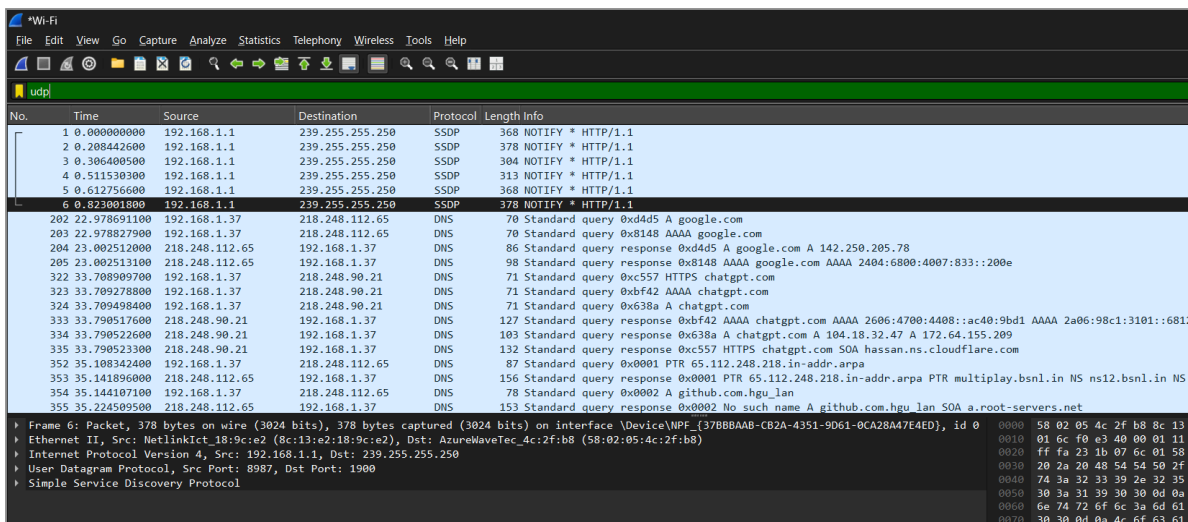


Figure 3.8: Applied UDP display filter

Step 9 — Deep Packet Inspection of Transport Layer Security Frame

Packet 14 was selected to examine detailed header fields: Frame (1132 bytes), Ethernet II, IPv6 (2001:4490:4ec1:1fb:ec30:ef21:70b:31e9, Dst: 2606:4700:4408::ac40:9bd1 under TLS.

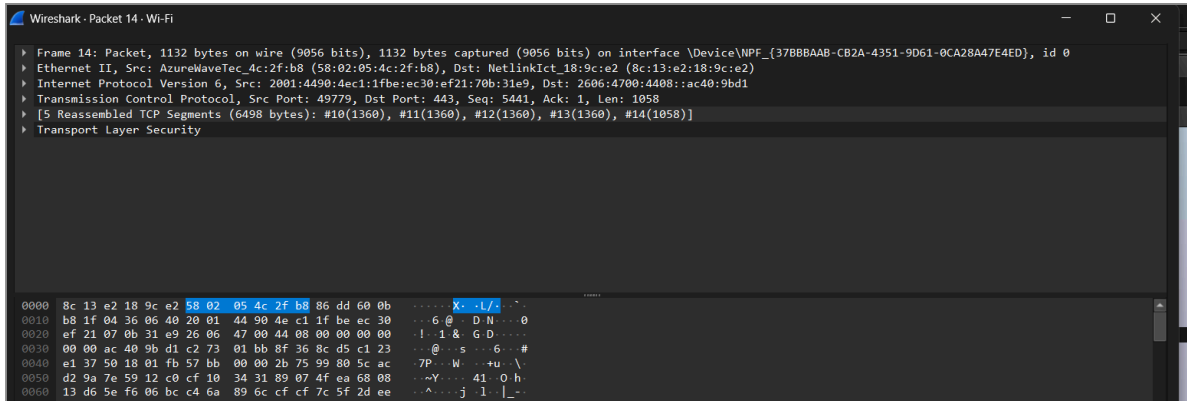


Figure 3.9: Deep packet inspection of TLS frame 14

Step 10 — Following TCP Stream Application Payload

The *Follow TCP Stream* feature (tcp.stream eq 1) was executed to inspect the raw data stream. The stream payload confirmed active cryptographic encryption preventing plaintext credential extraction.

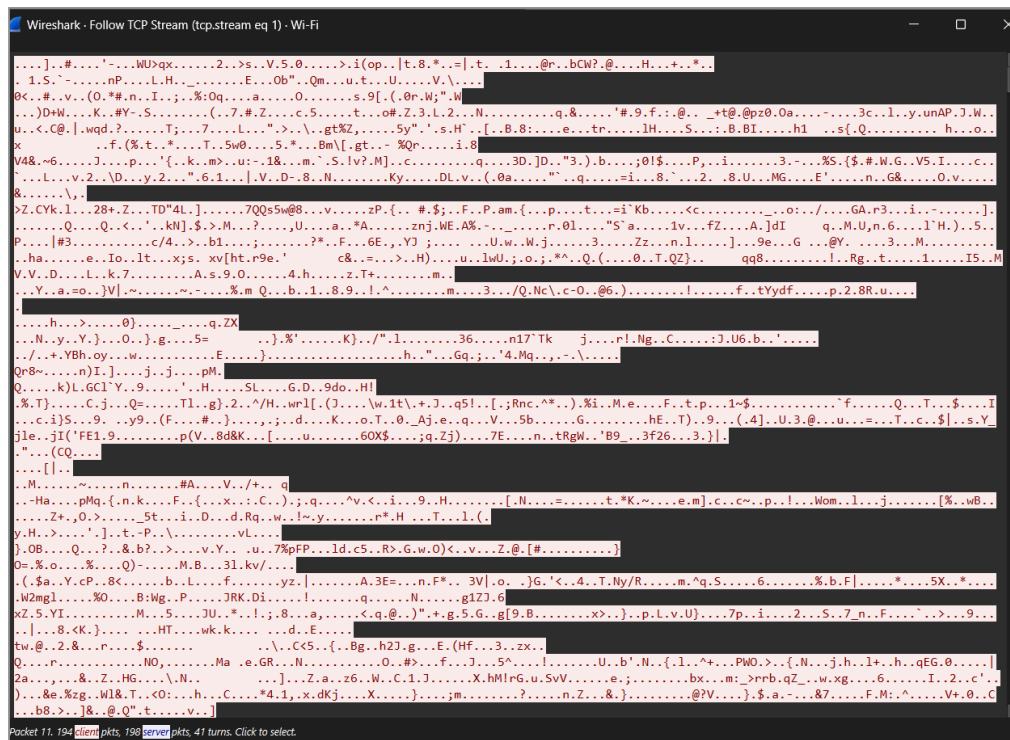
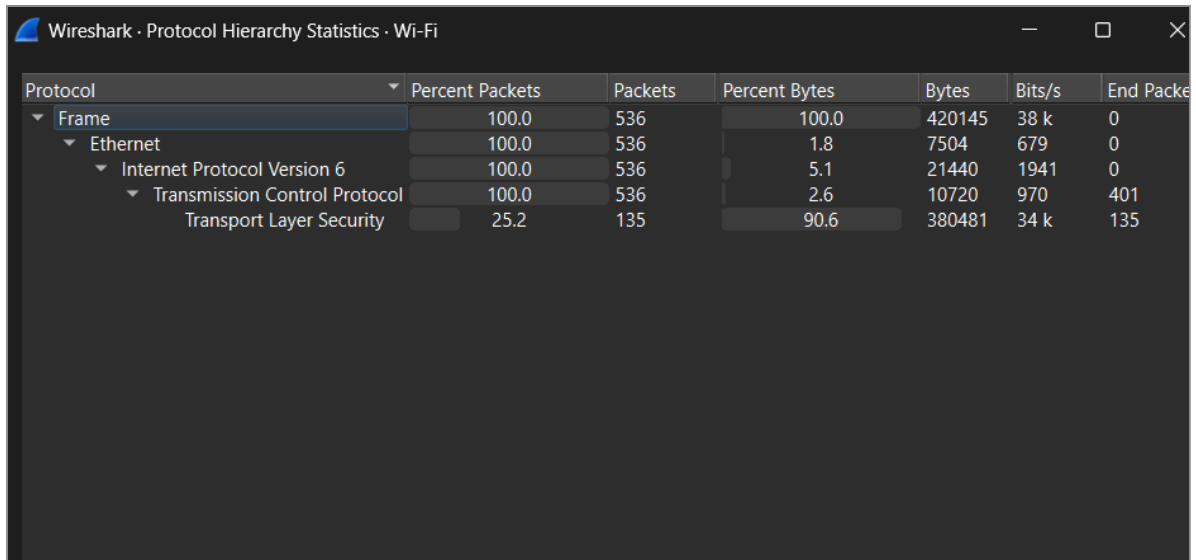


Figure 3.10: Follow TCP Stream payload showing encrypted session data

Step 11 — Protocol Hierarchy Statistics Analysis

The *Protocol Hierarchy Statistics* window was opened, detailing packet distribution: 100% Ethernet/IPv6/TCP traffic, with Transport Layer Security (TLS) accounting for 90.6% of transmitted bytes.

The image shows the 'Wireshark - Protocol Hierarchy Statistics - Wi-Fi' window. It displays a tree view of protocols on the left and a corresponding table of statistics on the right. The tree view shows 'Frame' expanded to 'Ethernet', which is expanded to 'Internet Protocol Version 6', which is expanded to 'Transmission Control Protocol', which is expanded to 'Transport Layer Security'. The table shows the following data:

Protocol	Percent Packets	Packets	Percent Bytes	Bytes	Bits/s	End Packet
Frame	100.0	536	100.0	420145	38 k	0
Ethernet	100.0	536	1.8	7504	679	0
Internet Protocol Version 6	100.0	536	5.1	21440	1941	0
Transmission Control Protocol	100.0	536	2.6	10720	970	401
Transport Layer Security	25.2	135	90.6	380481	34 k	135

Figure 3.11: Protocol hierarchy statistics distribution window

OBSERVATIONS

1. 755 total packets were captured across the active Wi-Fi interface.
2. DNS resolution traffic tracked standard queries and responses for `google.com`, `chatgpt.com`, and `github.com`.
3. Transport Layer Security (TLS) represented 90.6% of network traffic payload.
4. Following TCP Stream displayed encrypted cryptographic ciphertexts rather than plaintext form inputs.

FINDINGS

- **Technical Protocol Evaluation:** The laboratory exercise evaluated packet traffic across network layers. In the empirical capture session, modern web traffic utilized HTTPS/TLS encryption over port 443. Consequently, data payloads were encrypted, preventing plaintext credential interception and demonstrating the forensic impact of transport layer encryption.
- **Protocol Analysis Capability:** Wireshark effectively dissected traffic across OSI layers 2 through 7 (Ethernet II -> IPv4/IPv6 -> TCP/UDP -> TLS/DNS).

RUBRICS

Criteria & Marks Assigned	Mark Allotted	Mark Awarded
1. GitHub Activity & Submission Regularity	3	
2. Application of Forensic Tools & Practical Execution	3	
3. Documentation & Reporting	2	
4. Engagement, Problem-Solving & Team Collaboration	2	
Total	10	

RESULT

Successfully Completed. Wireshark successfully captured live network packets, filtered protocols across layers 3–7, analyzed stream payloads, and generated protocol hierarchy statistics.

CONCLUSION

Wireshark captured and analyzed live network traffic across multiple protocols. Inspection of the TCP streams and protocol hierarchy statistics demonstrated that active TLS encryption protected application payloads from plaintext credential sniffing.

REFERENCES

- Wireshark User's Guide (Wireshark Foundation)
- Digital Forensics Laboratory Manual: Password Capturing

EXPERIMENT 04

Analyze Email Headers and Detect Email Spoofing Using MHA (Mail Header Analyzer)

OBJECTIVE

To access, extract, and analyze raw email internet headers, trace routing MTA servers via **Received:** fields, verify email authentication mechanisms (SPF, DKIM, DMARC), and evaluate potential email spoofing.

TOOLS / REQUIREMENTS

- **Email Platform / Client:** Gmail Web Interface
- **Analysis Tools:** Text Editor (Notepad) / Visual Email Header Inspection
- **Online Verification Utilities:** IP Geolocation & WHOIS Lookup (WhatIsMyIP / LookIP.net)

EXPERIMENT SCENARIO

Controlled laboratory evidence was used for this experiment.

An institutional email communication was analyzed to determine authenticity, trace transmission infrastructure, and identify potential spoofing indicators. The investigator accessed the RFC 822 / MIME header, parsed MTA intermediate routing hops, executed IP lookup on origin servers, and verified cryptographic SPF, DKIM, and DMARC alignment.

EVIDENCE / INPUT

- **Target Evidence File:** GATE 2027 & JAM 2027 Examinations - Reg.eml
- **Sender Address:** "Dr. R. Raja Subramanian HOD-CSE" <hodcse@klu.ac.in>
- **Recipient Address:** csegrp@klu.ac.in, 2024cse@klu.ac.in, 2023cse@klu.ac.in, 2025cse@klu.ac.in
- **Message-ID:** <CAMq6mVb2KZH-MSBe-cAKSiLSC1EU41MSv0o6e5-j5=c0bjpLXg@mail.gmail.com>
- **Delivery Time:** Tue, Aug 25, 2026 at 6:31 PM (Delivered after 23 seconds)
- **Sending MTA Host IP:** 209.85.220.69 / 209.85.220.41

PROCEDURE

Step 1 — Accessing the Email in Mail Client

The target email ("Fwd: GATE 2027 & JAM 2027 Examinations - Reg") was opened in the Gmail interface to inspect sender headers, delivery details, and message context.

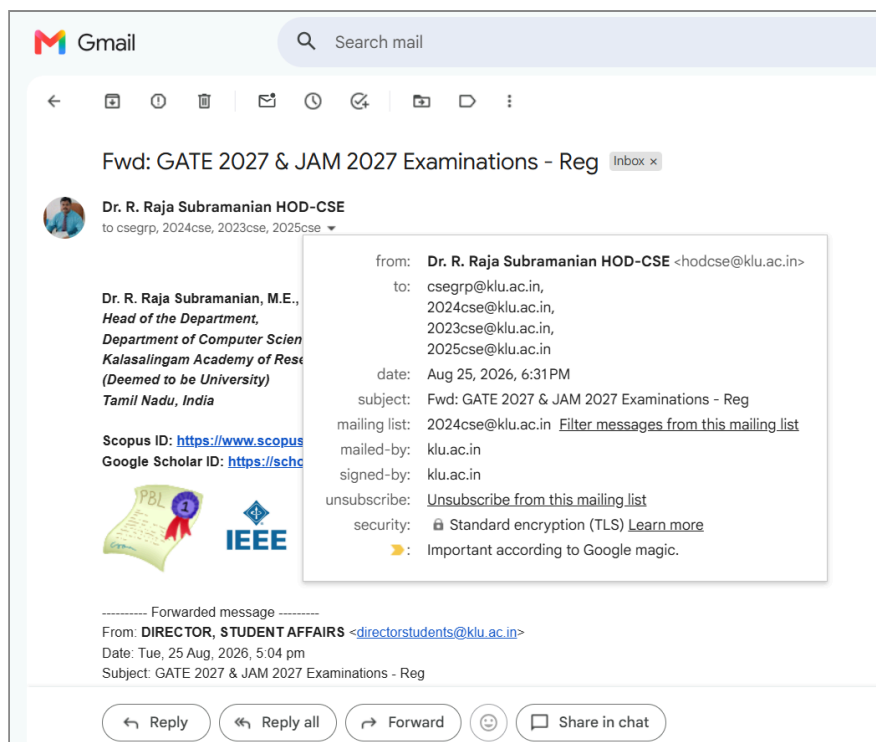


Figure 4.1: Target email accessed in mail client

Step 2 — Viewing Original Message Metadata

The *Show original* option was selected from the message menu, opening the RFC header overview displaying Message ID, creation timestamp, From, and To fields.

Original Message	
Message ID	<CAMq6mVb2KZH-MSBe-cAKSiLSC1EU41MSv0o6e5-j5=cObj
Created at:	Tue, Aug 25, 2026 at 6:31 PM (Delivered after 23 seconds)
From:	"Dr. R. Raja Subramanian HOD-CSE" <hodcse@klu.ac.in>
To:	csegrp@klu.ac.in, 2024cse@klu.ac.in, 2023cse@klu.ac.in, 2025

Figure 4.2: Original message metadata overview

Step 3 — Inspecting Mail Authentication Overview

The authentication overview table was evaluated:

Authentication Check	Evaluation Status	Details
SPF	PASS	Permitted sender IP 209.85.220.69
DKIM	PASS	Domain klu.ac.in signature verified
DMARC	PASS	Policy compliant

Original Message	
Message ID	<CAMq6mVb2KZH-MSBe-cAKSiLSC1EU41MSv0o6e5-j5=cObjpLXg@mail.gmail.com>
Created at:	Tue, Aug 25, 2026 at 6:31 PM (Delivered after 23 seconds)
From:	"Dr. R. Raja Subramanian HOD-CSE" <hodcse@klu.ac.in>
To:	csegrp@klu.ac.in, 2024cse@klu.ac.in, 2023cse@klu.ac.in, 2025cse@klu.ac.in
Subject:	Fwd: GATE 2027 & JAM 2027 Examinations - Reg
SPF:	PASS with IP 209.85.220.69 Learn more
DKIM:	'PASS' with domain klu.ac.in Learn more
DMARC:	'PASS' Learn more

Figure 4.3: Email authentication status table (SPF, DKIM, DMARC)

Step 4 — Extracting Full RFC 822 Raw Headers

The complete original raw MIME header was displayed with *Download Original* and *Copy to clipboard* controls for detailed offline forensic parsing.

The screenshot shows the Gmail 'Original Message' interface. At the top, there's a table with email metadata: Message ID, Created at, From, To, Subject, SPF, DKIM, and DMARC. Below this table, there are two buttons: 'Download Original' and 'Copy to clipboard'. At the bottom of the interface, the full RFC 822 raw MIME headers are displayed in a text box. The headers include: Delivered-To, Received, X-Forwarded-Encrypted, X-Received, ARC-Seal, and b=.

```
Delivered-To: 99249041233@klu.ac.in
Received: by 2002:a05:7300:e113:b0:315:b785:cfbb with SMTP id hd19csp4359716dyb; Tue, 25 Aug 2026 06:01:36 -0700 (PDT)
X-Forwarded-Encrypted: i=5; AHgh+RrscZupK0t1JLANS719ImE2lRPGAB5L0nFlzddL8Pz04UVO1qdYKCuvjHk1dH8bIVgdsAU2AcZCaUxg/w=@klu.ac.in
X-Received: by 2002:a05:6402:505b:b0:698:af31:5a9b with SMTP id 4fb4d7f45d1cf-6a42f1e08c0mr38071921a12.9.1787662895622; Tue, 25 Aug 2026 06:01:35 -0700 (PDT)
ARC-Seal: i=4; a=rsa-sha256; t=1787662895; cv=pass; d=google.com; s=arc-20260327; b=M4SEaA7xG61E0GExP2Ho13mduYKPMgBxclMv4-Tu02THFdh51g8akKwuv5B71u0K
```

Figure 4.4: Full original RFC 822 MIME headers interface

Step 5 — Analyzing Primary Receiving MTA Hop

The raw header was pasted into a text editor (email-header.txt). The top Received: line (Received: by 2002:a05:7300:e113:b0:315:b785:cfb6 with SMTP id hd19csp4359716dyb; Tue, 25 Aug 2026 06:01:36 -0700 (PDT)) was examined to confirm the final delivering MTA and ARC security seal.

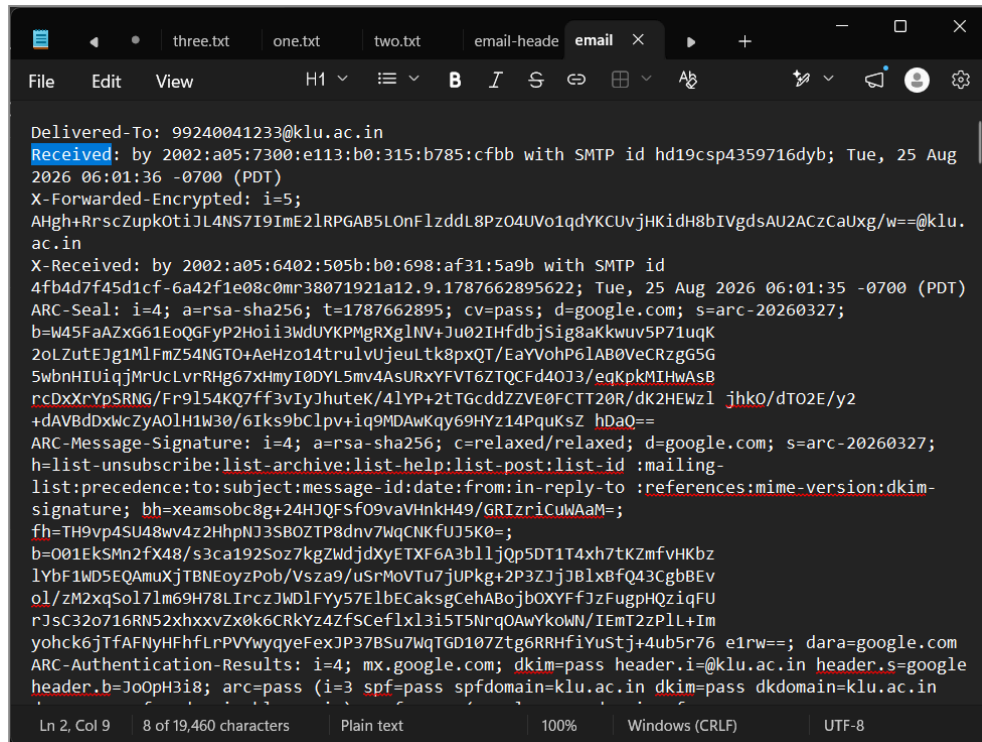


Figure 4.5: Examination of top Received header hop

Step 6 — Reviewing Comprehensive Header Stream

The entire structured header stream was analyzed to verify the chronological relay sequence across intermediate mail transfer agents.

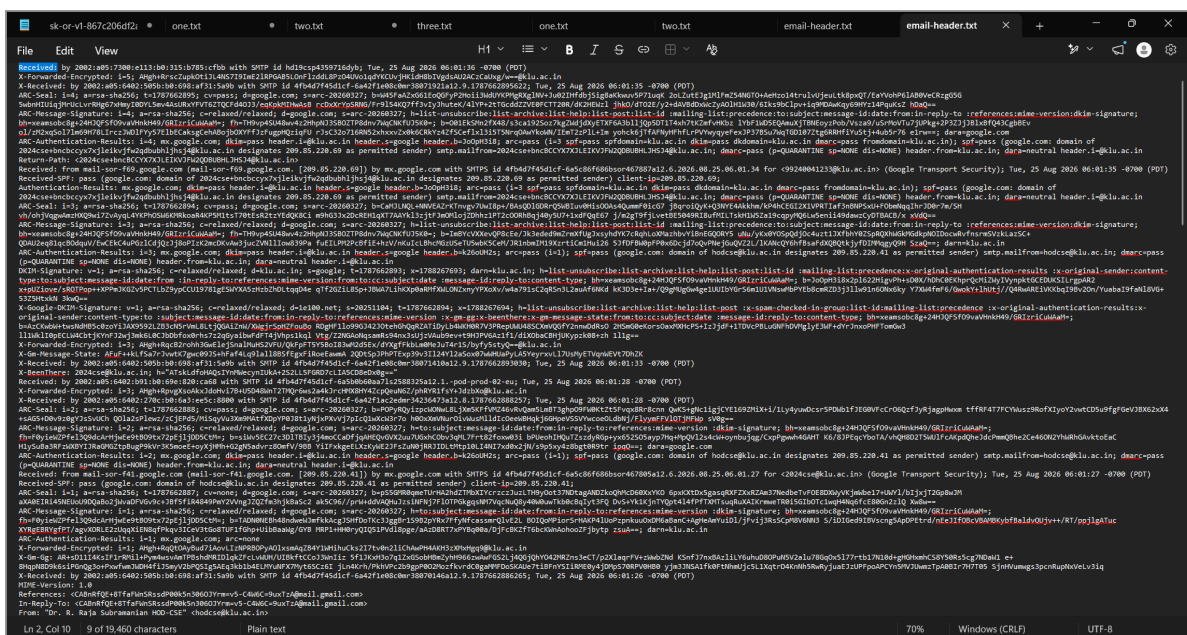


Figure 4.6: Complete raw email headers analysis

Step 7 — Detailed Inspection of Initial SMTP Hop

A focused examination of the initial receiving hop was conducted to verify protocol compliance (SMTP id hd19csp4359716dyb; Tue, 25 Aug 2026 06:01:36 -0700).

[illegible]

Figure 4.7: Zoomed inspection of initial SMTP receiving hop

Step 8 — Identifying Key Header Fields

Key forensic metadata fields were cataloged from the header:

- From: "Dr. R. Raja Subramanian HOD-CSE" <hodcse@klu.ac.in>
- Subject: Fwd: GATE 2027 & JAM 2027 Examinations - Reg
- Date: Tue, 25 Aug 2026 18:31:12 +0530
- Message-ID: <CAMq6mVb2KZH-MSBe-cAKSiLSC1EU41MSv0o6e5-j5=c0bjpLXg@mail.gmail.com>

[illegible]

Figure 4.8: Identification of key RFC header fields

Step 9 — Geolocation and WHOIS IP Lookup of Sending Server

The sending relay IP address 209.85.220.69 was queried through an online IP lookup service, confirming:

- **Owner / ISP:** Google LLC
- **Autonomous System:** AS15169
- **CIDR Block:** 209.85.208.0/20
- **Location:** Utica, New York, United States

The IP address 209.85.220.69 is owned by Google LLC and is located in Utica, New York, United States.

Ownership and Network Details

- **Owner/ISP:** Google LLC
- **ASN (Autonomous System Number):** AS15169
- **CIDR Block:** 209.85.208.0/20 (some sources list 209.85.192.0/18)
- **IP Type:** Public IPv4
- **Hostname:** Associated with Google infrastructure, no specific websites detected running directly on this IP [LookIP.net +3](#) .

Geolocation

- **City:** Utica
- **State:** New York
- **Country:** United States
- **Time Zone:** Eastern Time Zone (Olson TZ database)
- **Accuracy:** Approximate; IP geolocation is based on ISP registration and routing information, not the exact physical location of a device [WhatIsMyIP.com +1](#) .

Figure 4.9: IP lookup and geolocation analysis for sending MTA

Step 10 — Evaluating Sender Policy Framework (SPF) Status

The SPF authentication record in the header was verified: `spf=pass` (google.com: domain of `hodcse@klu.ac.in` designates 209.85.220.41 as permitted sender).

```
Content-Type: multipart/mixed; boundary="00000000000050dbb40659deb328"
-Original-Sender: hodcse@klu.ac.in
-Original-Authentication-Results: mx.google.com;
    dkim=pass header.i=@klu.ac.in header.s=google header.b=k26oUH2s;
    arc=pass (i=1); spf=pass (google.com: domain of hodcse@klu.ac.in designates 209.85.220.41 as permitted sender);
    p=QUARANTINE sp=NONE dis=NONE) header.from=klu.ac.in; dara=neutral header.i=@klu.ac.in
Precedence: list
mailing-list: list 2024cse@klu.ac.in; contact 2024cse+owners@klu.ac.in
List-ID: <2024cse.klu.ac.in>
```

Figure 4.10: Sender Policy Framework (SPF) verification in raw header

Step 11 — Verifying DomainKeys Identified Mail (DKIM) Signature

The DKIM cryptographic signature header was evaluated: DKIM-Signature: v=1; a=rsa-sha256; c=relaxed/relaxed; d=klu.ac.in; s=google; confirming signature validity and body hash integrity.

```
ARC-Authentication-Results: i=3; mx.google.com; dkim=pass header.i=@klu.ac.in header.s=google header.b=k26oUH2s; arc=pass (i=1); spf=pass (google.co
hodcse@klu.ac.in designates 209.85.220.41 as permitted sender) smtp.mailfrom=hodcse@klu.ac.in; dmarc=pass (p=QUARANTINE sp=NONE dis=NONE) header.fro
dana=neutral header.i=@klu.ac.in
DKIM-Signature: v=1; a=rsa-sha256; c=relaxed/relaxed; d=klu.ac.in; s=google; t=1787662893; x=1788267693; darn=klu.ac.in; h=list-unsubscribe:list-arc
help:list-post:list-id :mailing-list:precedence:x-original-authentication-results :x-original-sender:content-type:to:subject:message-id:date:from :i
to:references:mime-version:from:to:cc:subject:date :message-id:reply-to:content-type; bh=xeamsobc8g+24HJQFSf09vaVHnkH49/GRIzriCuWAAm=;
b=JoOpH3i8x2p1622HigvPhsD0X/hDhC0EKhrQcMiZWyIVynpKtGCEUDUKSILrnpAR2 x+pUzIove/sRQTPop++XPPmJKGZv5PCTLbZ9ypCCU19781gESWYXA5zHzbZDLtqqD4e
```

Figure 4.11: DKIM cryptographic signature header inspection

Step 12 — Examining Message-ID and ARC Authentication Seal

The ARC (Authenticated Received Chain) signature and message-id header references were analyzed to confirm end-to-end chain verification across forwarding hops.

```
ARC-Seal: i=3; a=rsa-sha256; t=1787662894; cv=pass; d=google.com; s=arc-20260327; b=CaMjLNQL+NNVEAZrKtNvgv7LwI8p+/BAsQD1GDRnQ5WBI0Hs00As4QumF8icG7
jBqro3QyK+Q3NYE4Aakhm/kP4hCEGI2X1VPRT1af3nBNP5XuF0bmLqg1nR3D0r7m/SH_vh/oh3VqguAmzhXQ9wi7ZvAyl4YKPhOSN6KMRkoaR4KPSM1ts170tEsR2tzYEaQK8Ci
9hG3Jx2DcREH1qXT7AAVkl3zjtfJmOw1oj2DhHz1PT2c0ORhBqj40ySUT+1xdFQqE67 j/m2gT9fjLvetBE5049RI8uFmILTskH1W5Za19capyWQ6Lw5enii49dazwCyDTBACB/x xVdQ==
ARC-Message-Signature: i=3; a=rsa-sha256; c=relaxed/relaxed; d=google.com; s=arc-20260327; h=list-unsubscribe:list-archive:list-help:list-post:list-id :mailing-
list:precedence:to:subject: message-id: date: from: in-reply-to : references: mime-version: dkim-signature; bh=xeamsobc8g+24HJQFSf09vaVHnkH49/GRIzriCuWAAm=;
fh=TH9vp4S4M8w4z2HhpNj3S80ZTP8dnv7nqCNKFUJ5K0; b=ImBYcVXXevQP8cEe/Jk3aded9mZrmXfUg1xshydYK7cRqHLoMazhbvYiBnEGQORY5
uNu/yKx0Y05pQdj0c4uzt1JXfbhY8Z5pRQXhMgKgdKpN0IDocwRvfnsrSVzklaz5C+ Q0AU2eq81gcB0dquV/EwCEkC4uPGz1CdJQzj38oPIzK2ncDKvAw3jucZVN1Iow839Pa
FuEILPM2PcBfiE+hzV/nKuIcLhCNgzUSaTUSwbK5CeM/JR1nbmIM19XzntiCm1Hui26 5JFDFBw0pFP0x6Dcdj7oQvPWejGuQVZ2L/JKANCQY6hf8sAfDx08Q0tkjyFDIWWagYQ9H Sza0==; darn=klu.ac.in
ARC-Authentication-Results: i=3; mx.google.com; dkim=pass header.i=@klu.ac.in header.s=google header.b=k26oUH2s; arc=pass (i=1); spf=pass (google.com; domain of
hodcse@klu.ac.in designates 209.85.220.41 as permitted sender) smtp.mailfrom=hodcse@klu.ac.in; dmarc=pass (p=QUARANTINE sp=NONE dis=NONE) header.from=klu.ac.in;
```

Figure 4.12: Message-ID and ARC authentication seal verification

OBSERVATIONS

1. Message traveled through authorized Google Workspace mail transfer agents in 23 seconds.
2. SPF authentication evaluated to PASS for sending IP 209.85.220.69 / 209.85.220.41.
3. DKIM cryptographic signature was signed by d=klu.ac.in and validated as PASS.
4. DMARC policy evaluated to PASS.
5. IP lookup for 209.85.220.69 verified ownership by Google LLC (AS15169).

FINDINGS

- **Authentication Alignment:** Complete alignment between the visible sender (hodcse@klu.ac.in), SPF permitted sender IP block, and DKIM cryptographic signature (d=klu.ac.in) verifies email authenticity.
- **Absence of Spoofing:** No anomalous routing hops, header modifications, or domain impersonation indicators were present.

RUBRICS

Criteria & Marks Assigned	Mark Allotted	Mark Awarded
1. GitHub Activity & Submission Regularity	3	
2. Application of Forensic Tools & Practical Execution	3	
3. Documentation & Reporting	2	
4. Engagement, Problem-Solving & Team Collaboration	2	
Total	10	

RESULT

Successfully Completed. The email header was extracted, parsed, and verified. SPF, DKIM, and DMARC checks confirmed message authenticity without spoofing.

CONCLUSION

Header analysis confirmed that the email originated from an authorized Google Workspace mail server for domain `klu.ac.in`. Cryptographic authentication mechanisms (SPF, DKIM, DMARC) verified header and content integrity.

REFERENCES

- RFC 5322 (Internet Message Format)
- RFC 7208 (Sender Policy Framework)
- RFC 6376 (DomainKeys Identified Mail)
- RFC 7489 (DMARC)
- Digital Forensics Laboratory Manual: Analyze email headers and detect email spoofing using MHA (Mail Header Analyzer)

EXPERIMENT 05

Use Autopsy to Create a Case and Import Evidence

OBJECTIVE

To create a digital forensics case in Autopsy, ingest and analyze storage evidence data sources, examine deleted files and metadata structures, and generate a comprehensive forensic report.

TOOLS / REQUIREMENTS

- **Forensic Platform:** Autopsy 4.23.1 (64-bit) / The Sleuth Kit (TSK) engine
- **Host Operating System:** Microsoft Windows 11
- **Evidence Source:** Virtual Hard Disk volume (VHD.vhd / New Volume (D:) 2.0 GB)

EXPERIMENT SCENARIO

Controlled laboratory evidence was used for this experiment.

A forensic investigation case was initiated in Autopsy to analyze a storage volume containing deleted artifacts. The examiner configured formal case information, ingested the data source, ran automated analysis modules, identified deleted credential remnants from unallocated space, inspected low-level Master File Table (MFT) records using Sleuth Kit `istat` integration, and generated an exportable forensic report.

EVIDENCE / INPUT

- **Case Name:** CASE-2026-04(AUTOPSY)
- **Case Number:** 04
- **Examiner Details:** SEKHAR REDDY (Phone: 9392746218, Email: sekharnandireddy338@gmail.com)
- **Ingested Data Source:** VHD.vhd_1 Host / New Volume (D:) 2.0 GB
- **Target Deleted File:** /img_VHD.vhd/vol_vol2/User_credentials.txt.txt (Size: 18 bytes, Unallocated, MFT Entry 45, Sequence 2)
- **Target Report Format:** Excel Spreadsheet (Excel.xlsx)

PROCEDURE

Step 1 — Configuring Case Information and Examiner Details

Autopsy 4.23.1 was launched and a new forensic case was initialized. In the **Optional Information** window, case details were configured: Case Number 04, Examiner Name SEKHAR REDDY, Phone 9392746218, Email sekharnandireddy338@gmail.com.

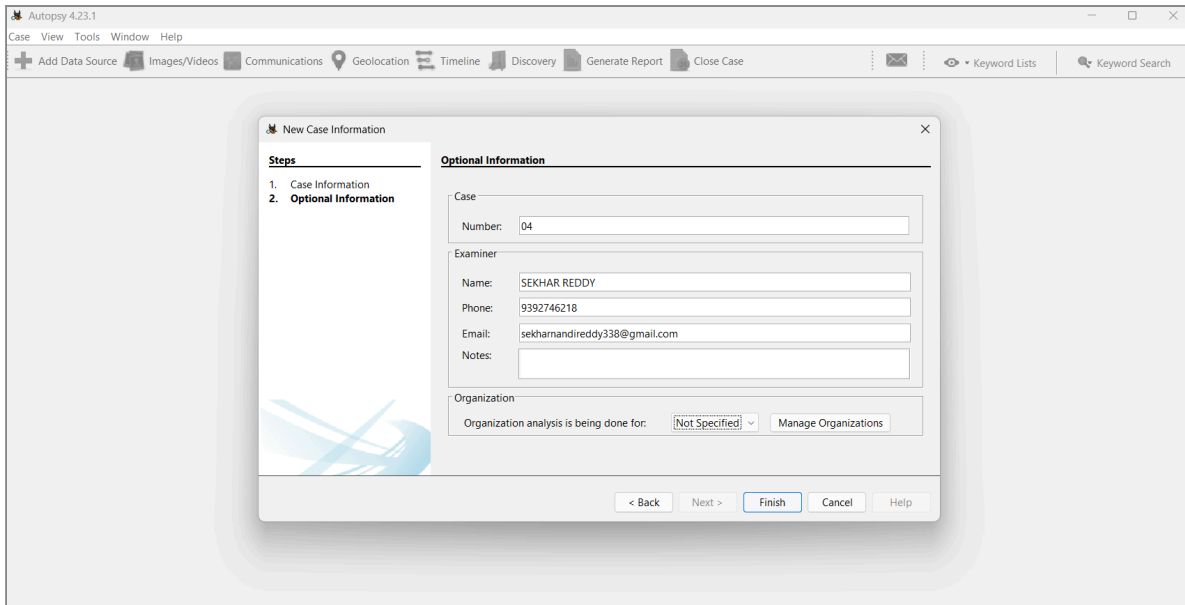


Figure 5.1: Autopsy new case optional information window

Step 2 — Selecting and Adding Storage Data Source

The *Add Data Source* wizard was executed. Under the Local Disk option, the target evidence drive New Volume (D:) 2.0 GB was selected with timezone set to (GMT+5:30) Asia/Calcutta.

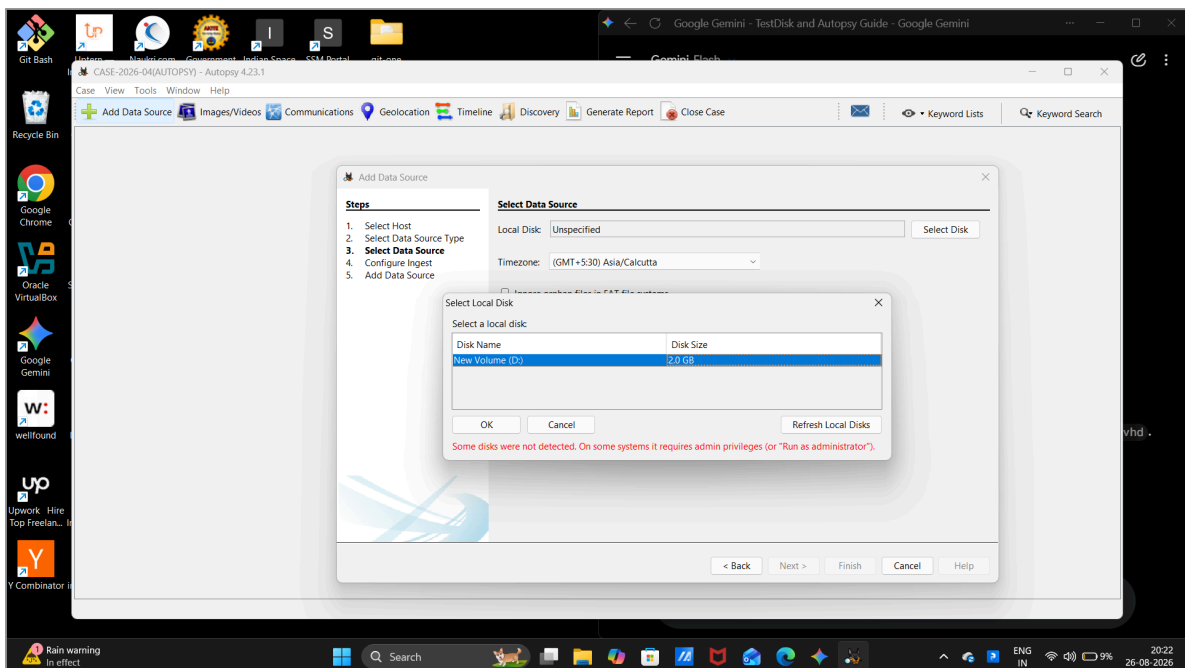


Figure 5.2: Adding local disk data source New Volume (D:)

Step 3 — Confirming Data Source Ingest into Local Database

The data source was committed and ingested into the SQLite case database, initiating automated ingest modules.

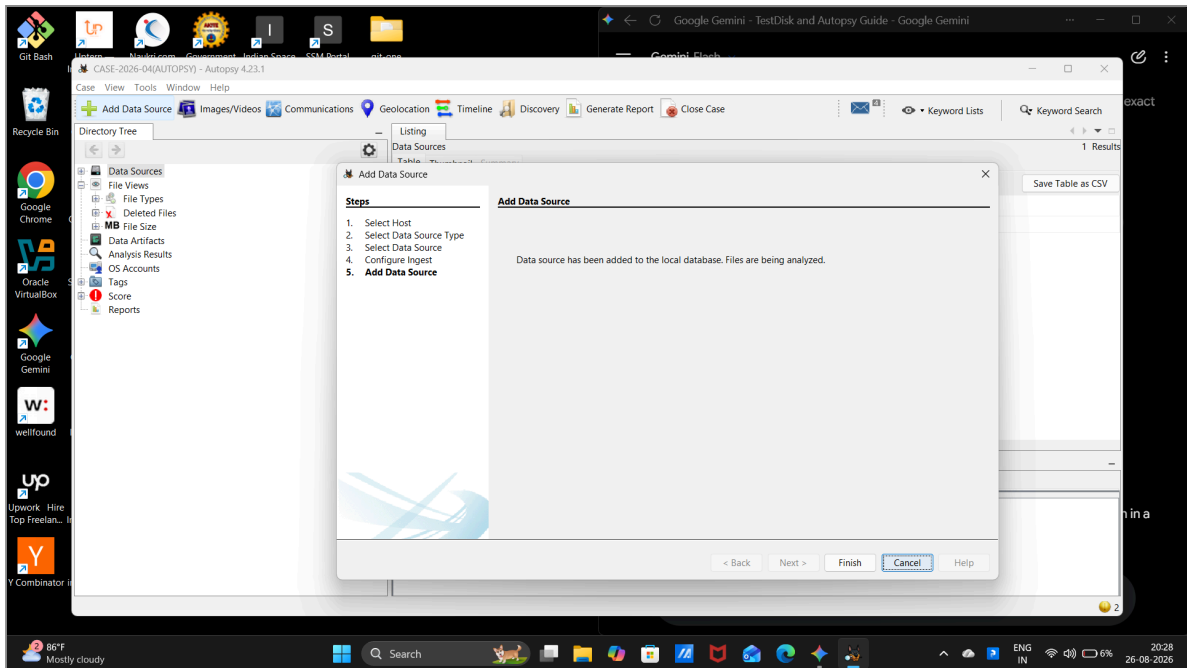


Figure 5.3: Data source ingest confirmation dialog

Step 4 — Exploring Directory Tree and Data Source Hierarchy

The Autopsy Directory Tree was navigated to inspect the ingested data source hierarchy, confirming active indexing of VHD.vhd_1 Host.

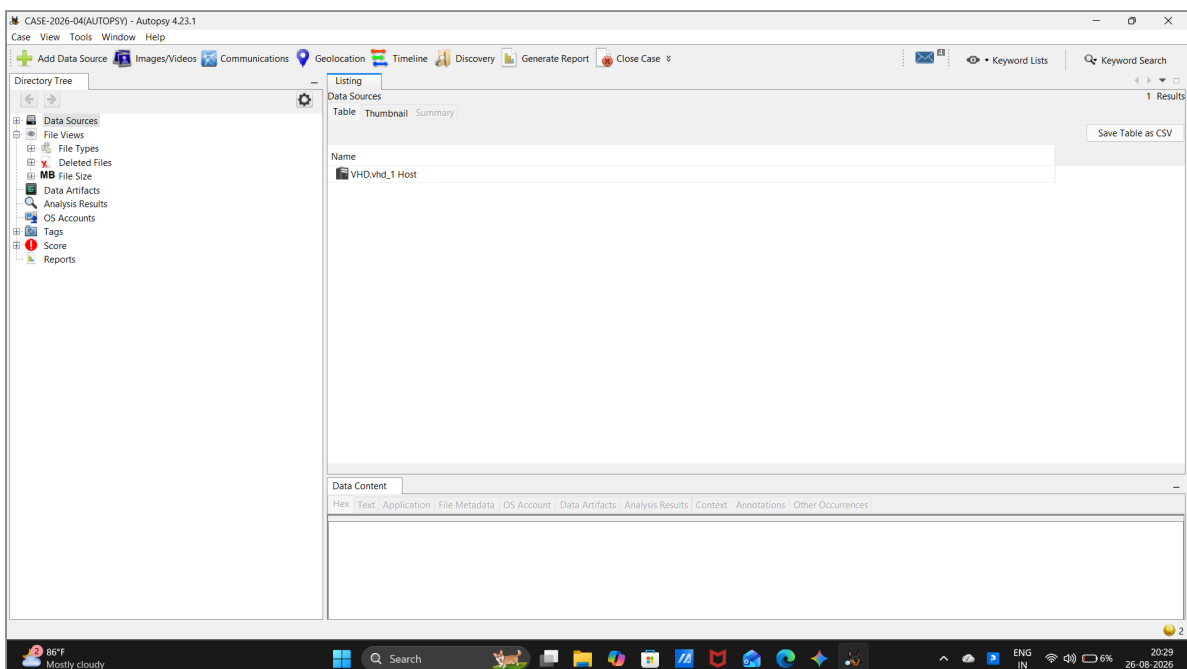


Figure 5.4: Directory tree displaying indexed data sources

Step 5 — Analyzing Deleted Files and Text Artifacts

Under **File Views -> Deleted Files -> All (1)**, the deleted artifact `User_credentials.txt.txt` (18 bytes, Unallocated) was identified. The lower Data Content pane (Text view) revealed the recovered credential payload: `admin:password@123`.

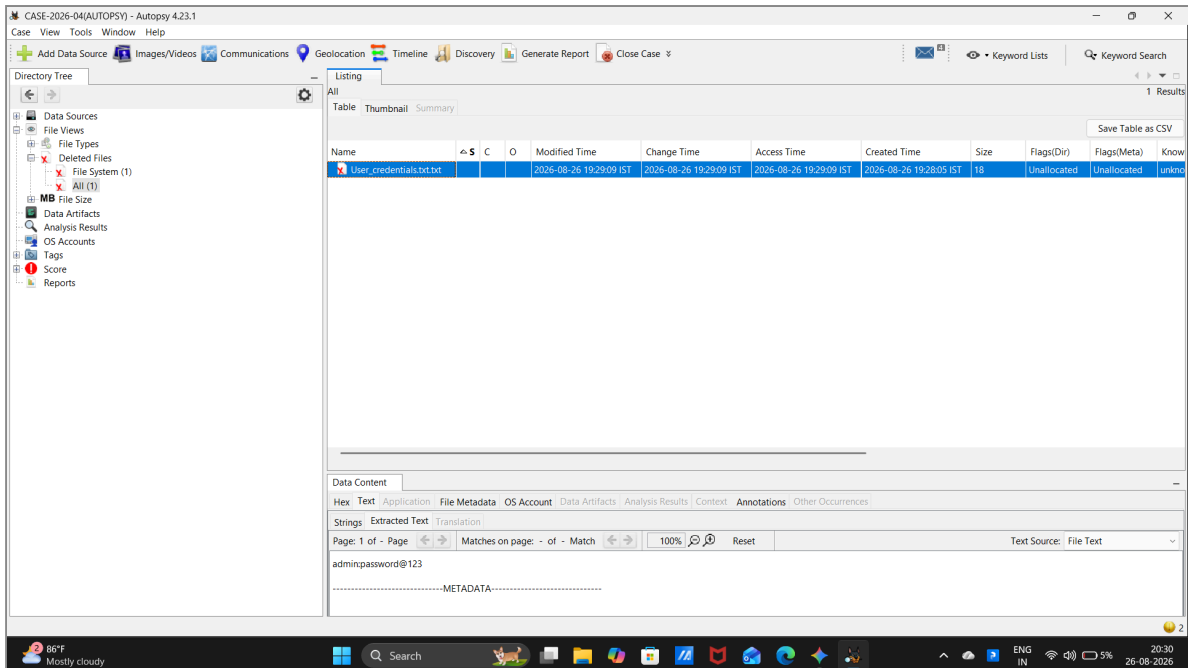


Figure 5.5: Deleted files view showing recovered text payload

Step 6 — Examining File Metadata and Sleuth Kit istat Output

The **File Metadata** tab was selected to analyze low-level filesystem attributes integrated from The Sleuth Kit **istat** tool:

- **File Name Allocation:** Unallocated
- **Metadata Allocation:** Unallocated
- **Modified / Accessed / Changed:** 2026-08-26 19:29:09 IST
- **Created:** 2026-08-26 19:28:05 IST
- **MFT Entry Header:** Entry: 45 Sequence: 2
- **Flags:** Archive

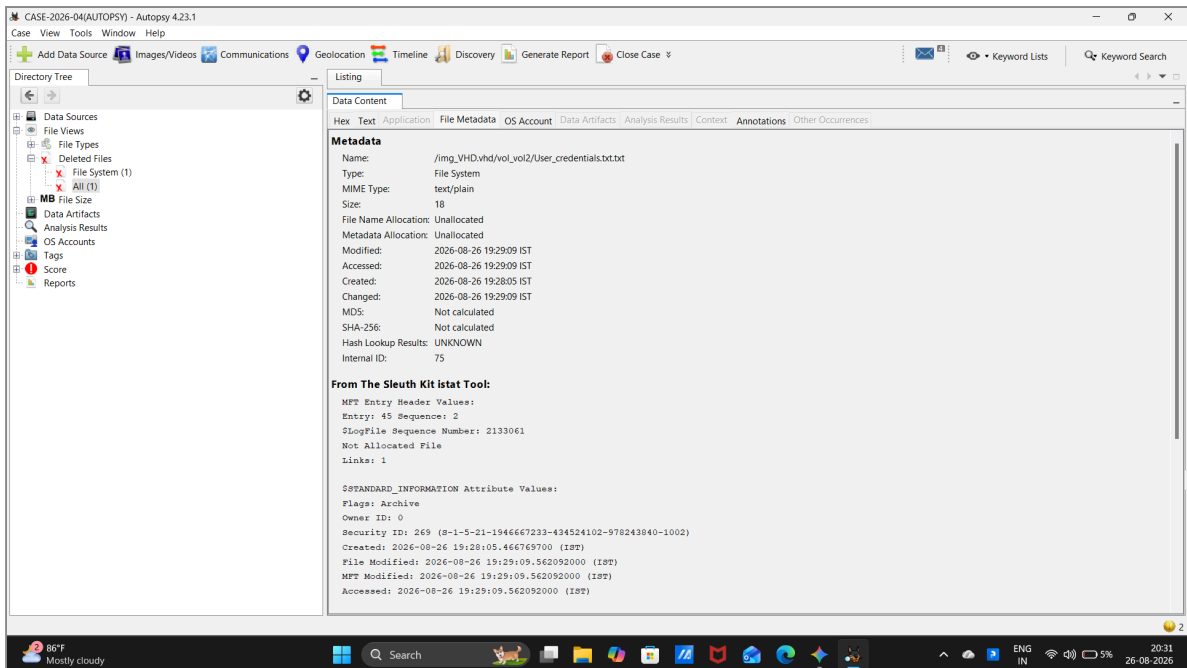


Figure 5.6: File metadata tab showing Sleuth Kit istat attributes

Step 7 — Configuring Forensic Report Generation Module

The *Generate Report* tool was opened from the toolbar. The **Excel Report** module was selected to create a structured spreadsheet report of findings and tagged items.

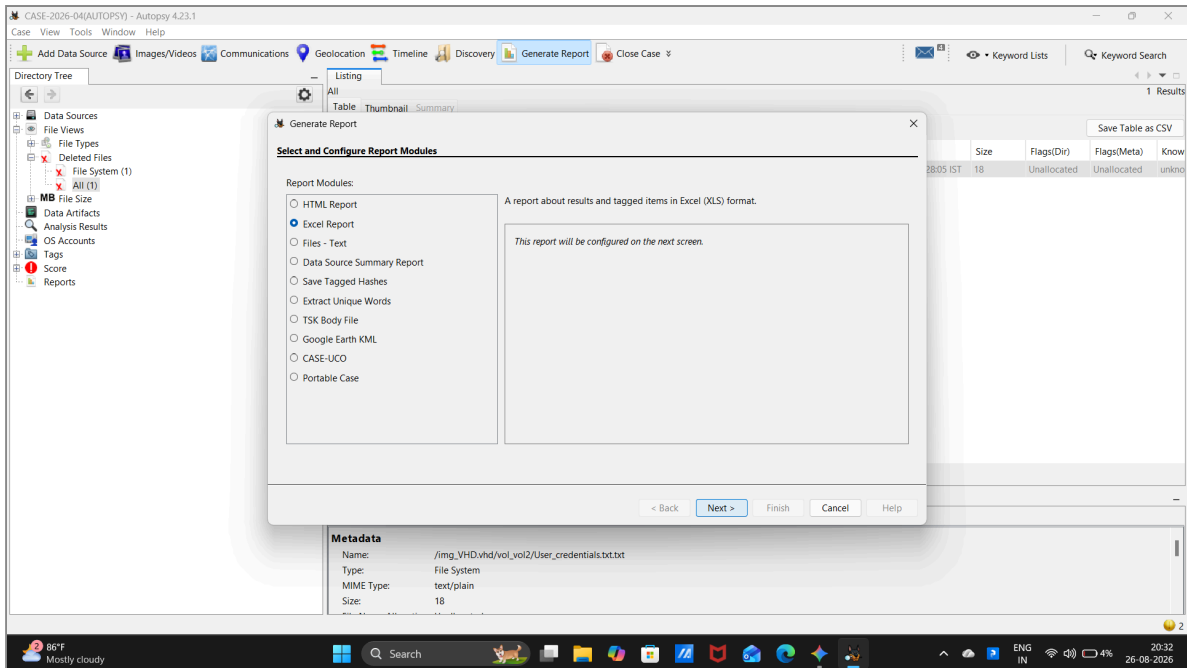


Figure 5.7: Generate Report dialog selecting Excel Report

Step 8 — Selecting Target Data Source for Reporting

In the report configuration wizard, the ingested data source VHD.vhd was selected for inclusion.

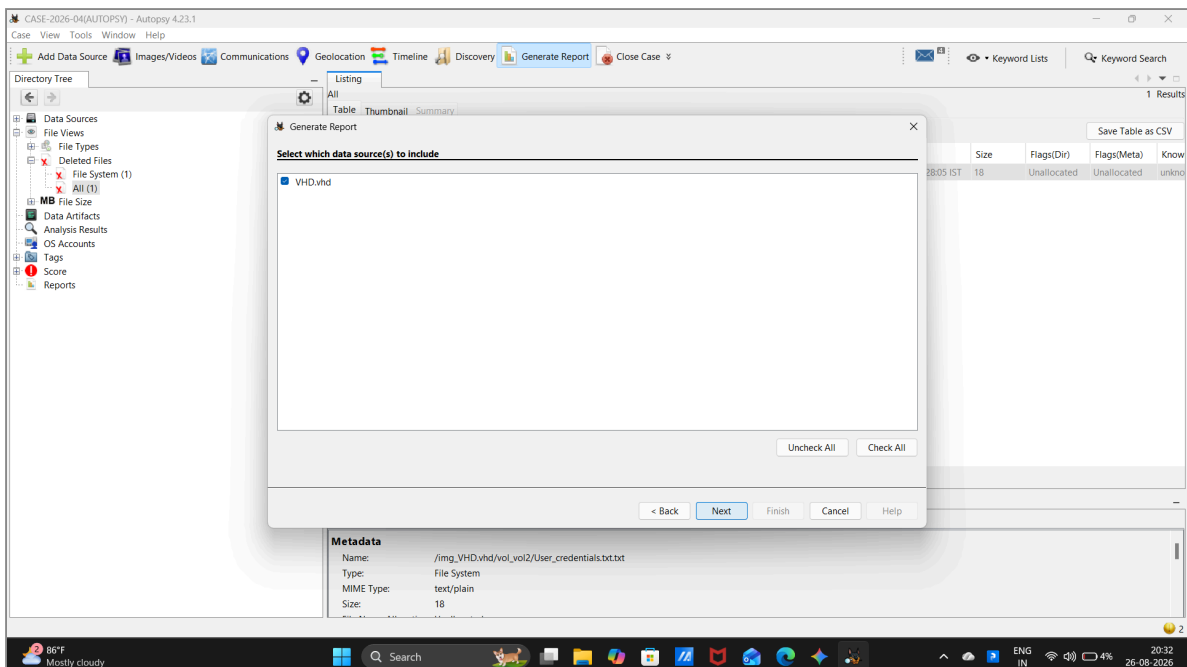


Figure 5.8: Selecting VHD data source for report inclusion

Step 9 — Executing Report Generation Pipeline

The report generation engine processed the case database and compiled artifact tables.

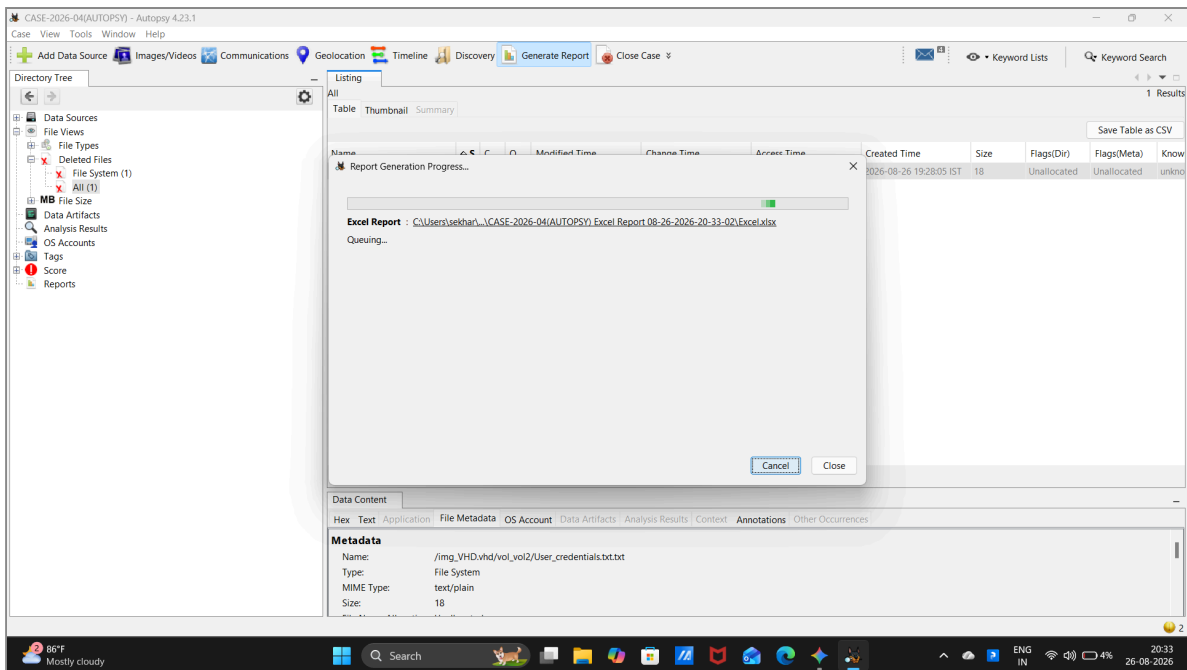


Figure 5.9: Report generation progress dialog

Step 10 — Verifying Final Report Generation Completion

The report generation pipeline concluded with status Complete. The generated report path was confirmed: CASE-2026-04(AUTOPSY) Excel Report 08-26-2026-20-33-02\Excel.xlsx.

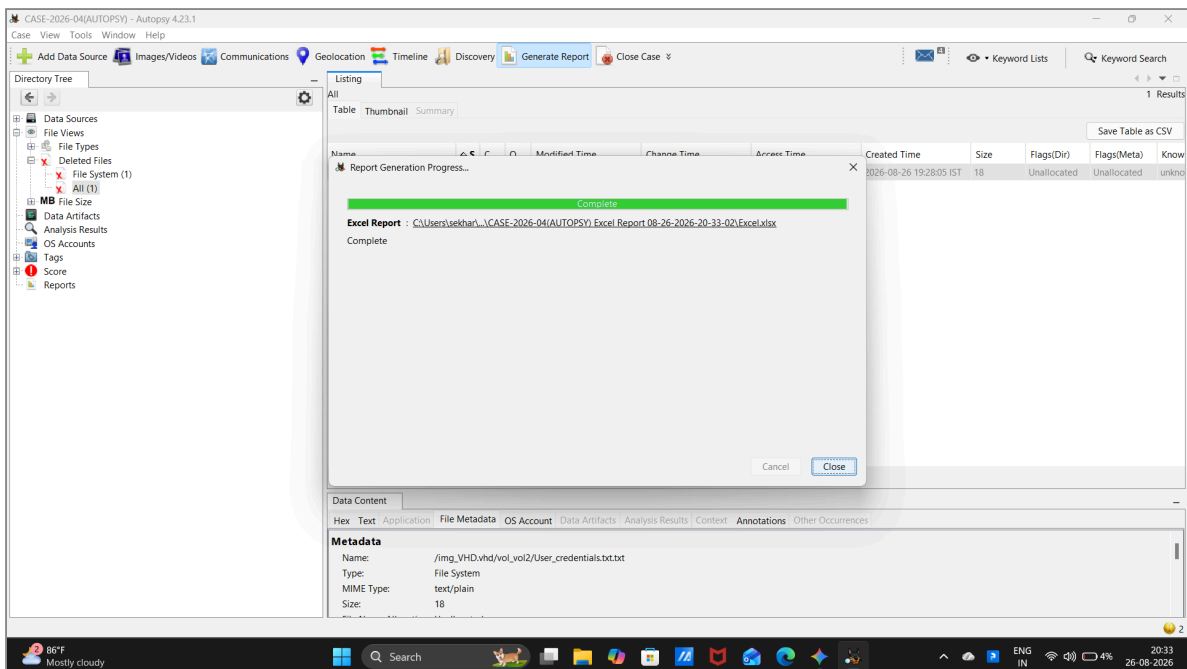


Figure 5.10: Report generation complete confirmation

OBSERVATIONS

1. Autopsy successfully created case CASE-2026-04(AUTOPSY) with examiner metadata.
2. Ingestion of data source VHD.vhd_1 Host indexed volume partitions and file structures.
3. Deleted file view isolated unallocated record User_credentials.txt.txt (18 bytes).

4. Content viewer extracted string `admin:password@123`.
5. Sleuth Kit `istat` metadata verified MFT Entry 45, Sequence 2, and timestamp records.
6. Report generation generated `Excel.xlsx` in the case reports directory.

FINDINGS

- **Artifact Carving and Recovery:** Autopsy parsed unallocated space on the NTFS volume to identify and preview deleted file contents without requiring full sector imaging.
- **Filesystem Metadata Auditing:** Integration with The Sleuth Kit provided low-level MFT inode and timestamp tracking essential for forensic reconstruction.
- **Reporting Soundness:** The automated report generation module compiled findings into a standardized audit format.

RUBRICS

Criteria & Marks Assigned	Mark Allotted	Mark Awarded
1. GitHub Activity & Submission Regularity	3	
2. Application of Forensic Tools & Practical Execution	3	
3. Documentation & Reporting	2	
4. Engagement, Problem-Solving & Team Collaboration	2	
Total	10	

RESULT

Successfully Completed. The forensic case was created in Autopsy, the disk image was ingested and analyzed, deleted credentials were recovered, and a comprehensive Excel report was generated.

CONCLUSION

Autopsy effectively managed the forensic case lifecycle from data source ingestion and unallocated space analysis to MFT metadata inspection and automated report compilation.

REFERENCES

- Autopsy User Documentation (Basis Technology / Sleuth Kit)
- Digital Forensics Laboratory Manual: Use Autopsy to create a case and import evidence